

Sprint 2 - Data analytics - Maria Palmer

📅 Date	September 21, 2026
⚙️ Status	Peer 2 peer Alejandro Pereyra
🏷️ Tags	SQL

Documentación

Important

Al llarg d'aquest sprint se't demanarà que facis alguns canvis a les taules, per exemple, afegint relacions, canviant els tipus de dades o modificant les *constraints*.

Tots aquests canvis els has de realitzar utilitzant codi SQL. No s'acceptaran solucions on els canvis hagin estat realitzats utilitzant el wizard.

Escriure els canvis en codi permet fer reproduïble el procés, i ser més àgil en futures modificacions.

Descripció

Repassar les nocions bàsiques per a l'ús de base de dades relacionals. En aquest sprint, iniciaràs la teva experiència pràctica amb una base de dades que conté informació d'una empresa dedicada a la venda de productes en línia. En aquesta activitat, t'enfocaràs en dades relacionades amb les transaccions efectuades i la informació corporativa de les empreses que van participar.



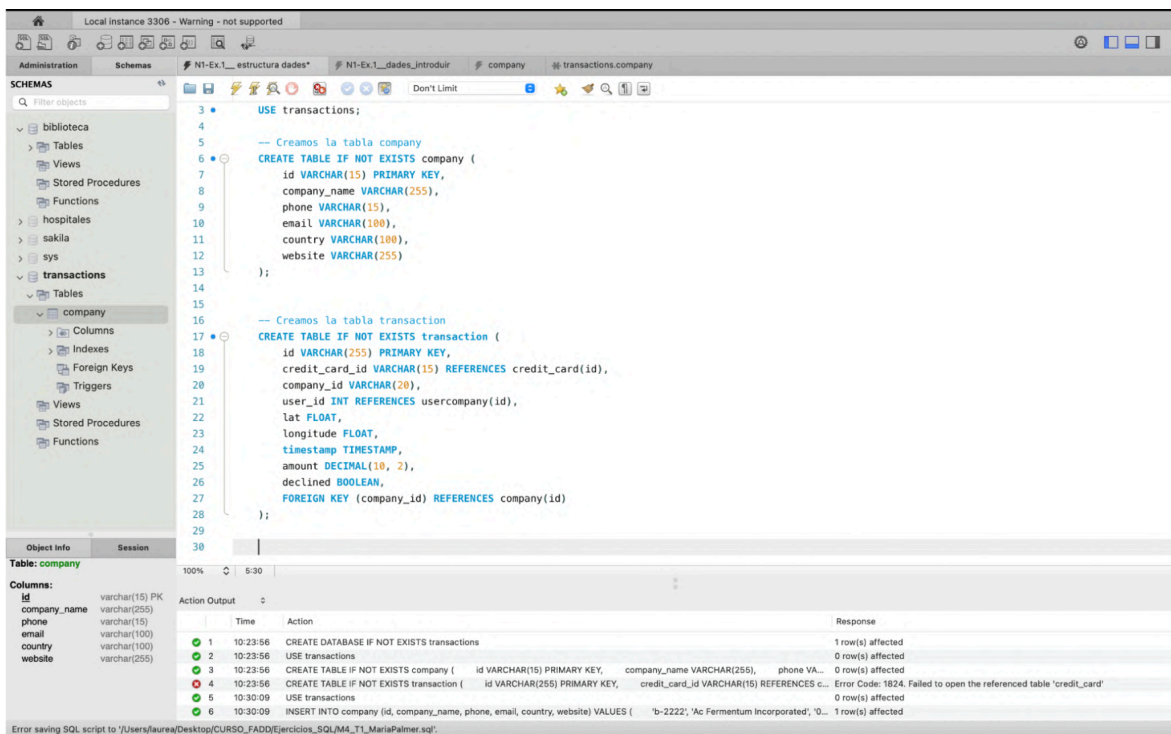
Important: Totes les transformacions i importacions que se't demanen en aquesta tasca s'han de realitzar utilitzant codi SQL. NO ES PERMET fer els canvis utilitzant el wizard.

▼ Nivell 1 ★

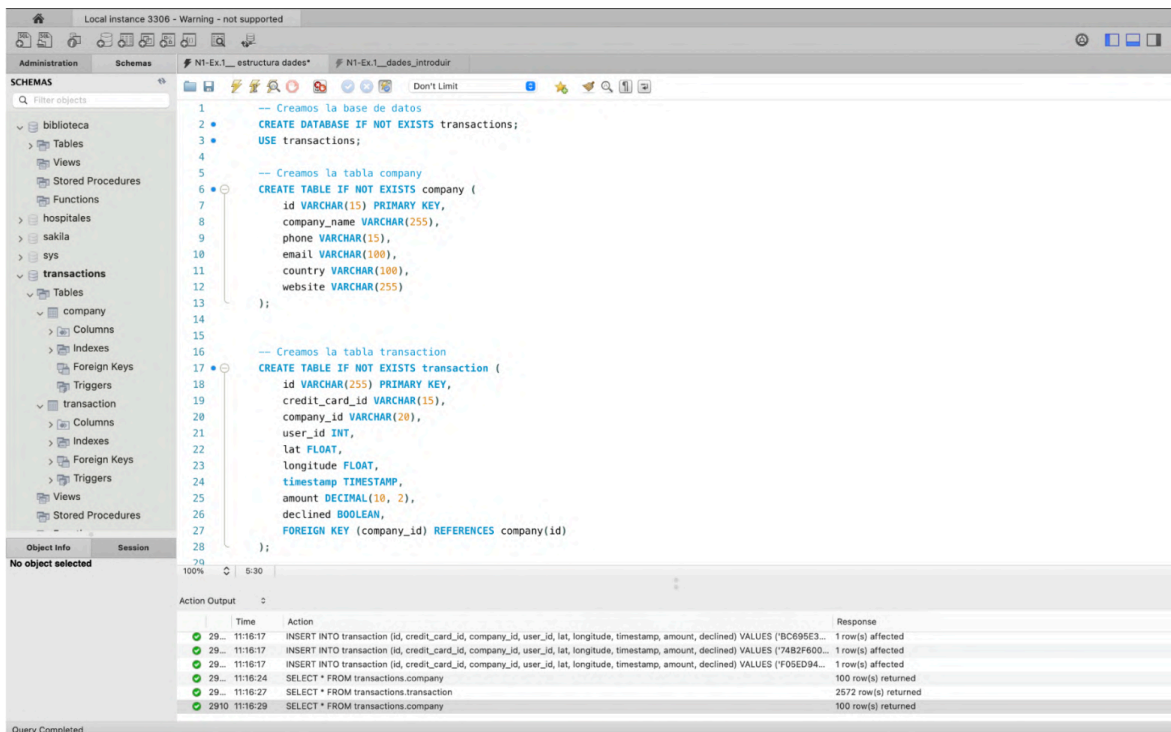
Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

Cuando ejecutamos los documentos adjuntos encontramos un error en la creación de una de las 2 tablas, cosa que nos impide crear correctamente la tabla "transaction". Esto se debe a que se está haciendo referencia a otras tablas que por ahora no hemos creado ni existen: credit_card y usercompany.



Lo que vamos a hacer por el momento es eliminar estas referencias para poder ejecutar correctamente esta parte del script y tener nuestra segunda tabla con los datos correctamente introducidos.



Después de solucionar el error tenemos nuestra base de datos "transactions" compuesta por dos tablas relacionadas llamadas "company" y "transaction". En este caso encontramos una relación 1:N donde **company** almacena los datos únicos (id) de las empresas como clave

primaria y se relaciona con todas las múltiples transacciones de transaction mediante la clave foránea (company_id).

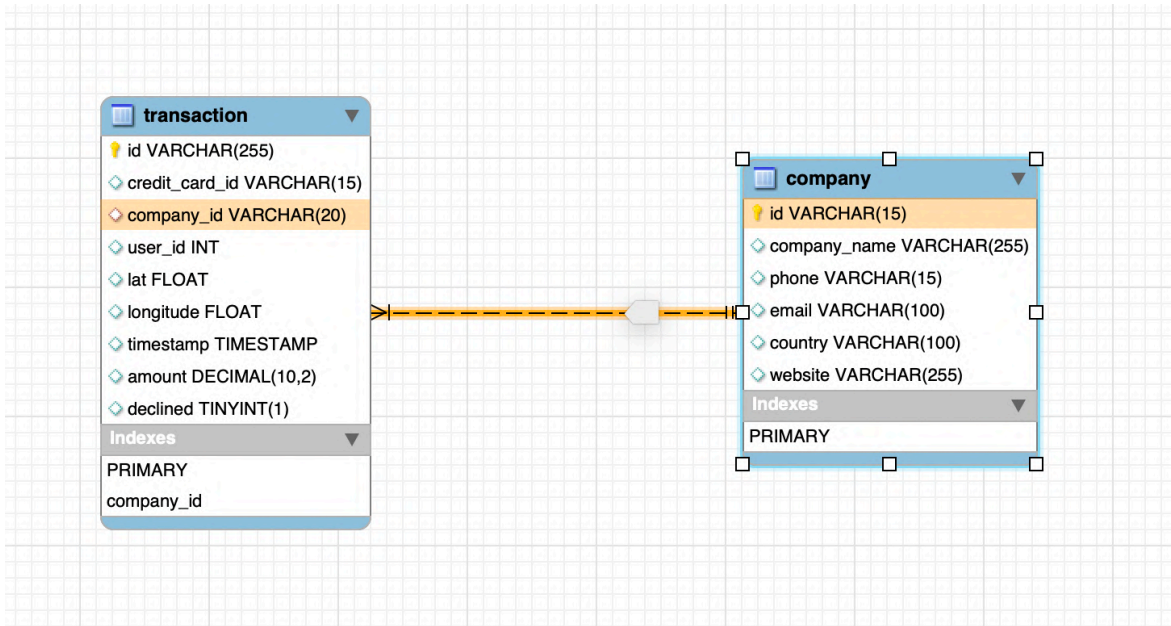


Table: **company**

Columns:

<u>id</u>	varchar(15) PK
company_name	varchar(255)
phone	varchar(15)
email	varchar(100)
country	varchar(100)
website	varchar(255)

Table: **transaction**

Columns:

id	varchar(255) PK
credit_card_id	varchar(15)
company_id	varchar(20)
user_id	int
lat	float
longitude	float
timestamp	timestamp
amount	decimal(10,2)
declined	tinyint(1)

Podemos observar el tipo de datos que recogen las dos tablas, donde mayormente encontraremos VARCHAR para campos alfanúmericos y/o con los que no se tiene que operar, INT para los números enteros, FLOAT y DECIMAL para los decimales y TINYINT para el boleano. También vemos indicada la clave primaria de cada tabla que corresponde al "id" de cada una.

Exercici 2

Utilitzant **JOIN** realitzaràs les següents consultes:

- Llistat dels països que estan generant vendes.

The screenshot displays a database management interface. At the top, a SQL query is entered in a text area:

```
1 • USE transactions;
2 -- SPRINT 2 - NIVELL 1
3 -- S2.N1.02 Utilitzant 'JOIN' realitzaràs les següents consultes:
4 -- Llistat dels països que estan generant vendes.
5
6 • SELECT country AS countries_selling
7 FROM company AS c
8 JOIN transaction AS t
9 ON c.id = t.company_id
10 WHERE t.company_id IS NOT null AND c.country IS NOT null AND t.declined = 0
11 GROUP BY country;
12
```

Below the query, the 'Result Grid' shows a list of countries. The first column is 'countries_selling'. The countries listed are: Netherlands, Sweden, New Zealand, Italy, Germany, United States, China, France, Spain, United Kingdom, Norway, Australia, Belgium, Ireland, and Canada.

At the bottom, the 'Action Output' tab shows the execution details of the query. It includes columns for 'Time', 'Action', 'Response', and 'Duration / Fetch Time'. The output shows that the query was executed successfully, returning 15 rows.

En la consulta original no había el condicional referente al país not null pero como he corregido esta consulta una vez ya hecho el ejercicio 6 (donde se añade un registro nuevo) pues se ve de este modo. En el condicional también se indica que solo se tengan en cuenta las transacciones que no han sido rechazadas (declined = 0). Esto se debe a que si no hay un intercambio real económico no se debería contar como venta efectiva.

En otro tipo de consultas se podrían tener en cuenta las transacciones rechazadas para resolución de problemas o atención al cliente, pero este no es el caso.

- Des de quants països es generen les vendes.

```

13 -- Des de quants països es generen les vendes.
14 • SELECT COUNT(DISTINCT country) AS total_countries_selling
15 FROM company AS c
16 JOIN transaction AS t
17 ON c.id = t.company_id
18 WHERE t.company_id IS NOT null AND c.country IS NOT null AND t.declined = 0;

```

total_countries_selling
15

Time	Action	Response	Duration / Fetch Time
01:28:29	SELECT pr.id, pr.product_name, COUNT(tp.product_id) AS purchase_amount FROM products_csv pr LEFT...	100 row(s) returned	0.173 sec / 0.000041...
01:29:47	SELECT pr.id, pr.product_name, COUNT(tp.product_id) AS purchase_amount FROM products_csv pr LEFT...	100 row(s) returned	0.173 sec / 0.000044...
01:31:22	SELECT pr.id, pr.product_name, COUNT(tp.product_id) AS purchase_amount FROM products_csv pr LEFT...	100 row(s) returned	0.173 sec / 0.000042...
11:45:34	SELECT iban, ROUND(AVG(amount), 2) AS avg_amount, company_name FROM transactions_csv t JOIN companies...	367 row(s) returned	0.044 sec / 0.000064...
11:47:22	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	Error Code: 1146. Table 'ecommerce.company' doesn't exist	0.0087 sec
11:47:32	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	16 row(s) returned	0.014 sec / 0.000014...
11:48:15	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	16 row(s) returned	0.0052 sec / 0.00000...
11:49:40	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	16 row(s) returned	0.013 sec / 0.000007...
11:50:18	SELECT country AS pais_veniendo FROM company AS c LEFT JOIN transaction AS t ON c.id = t.company_id WH...	16 row(s) returned	0.012 sec / 0.000009...
11:52:46	SELECT company_name, ROUND(AVG(amount), 2) AS top_ventas FROM company AS c JOIN transaction AS t ON c.i...	Error Code: 1054. Unknown column 'declined' in 'having clause'	0.0011 sec
11:55:05	SELECT company_name, AVG(amount) AS top_ventas FROM company AS c JOIN transaction AS t ON c.id = t.comp...	1 row(s) returned	0.013 sec / 0.000017...
11:55:28	SELECT company_name, ROUND(AVG(amount), 2) AS top_ventas FROM company AS c JOIN transaction AS t ON c.i...	1 row(s) returned	0.012 sec / 0.000006...
12:13:27	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	16 row(s) returned	0.013 sec / 0.000009...
12:13:53	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	0 row(s) returned	0.0052 sec / 0.00000...
12:14:30	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	0 row(s) returned	0.0041 sec / 0.00000...
12:16:10	SELECT country AS pais_veniendo FROM company AS c JOIN transaction AS t ON c.id = t.company_id WHERE t...	15 row(s) returned	0.0086 sec / 0.0000...
12:23:37	SELECT COUNT(DISTINCT country) AS total_countries_selling FROM company AS c JOIN transaction AS t ON c.i...	1 row(s) returned	0.013 sec / 0.000006...
12:24:29	SELECT COUNT(DISTINCT country) AS total_countries_selling FROM company AS c JOIN transaction AS t ON c.i...	1 row(s) returned	0.013 sec / 0.000011...

En este caso pasa lo mismo que en el anterior ejercicio con el tema de country not null.

- Identifica la companyia amb la mitjana més gran de vendes.

```

20 -- Identifica la companyia amb la mitjana més gran de vendes.
21 • SELECT company_name, ROUND(AVG(amount), 2) AS top_ventas
22 FROM company AS c
23 JOIN transaction AS t
24 ON c.id = t.company_id
25 WHERE declined = 0
26 GROUP BY company_name
27 ORDER BY 2 DESC
28 LIMIT 1;

```

company_name	top_ventas
Mattis Foundation	346.38

Time	Action	Response	Duration / Fetch Time
11:50:18	SELECT country AS pais_veniendo FROM company AS c LEFT JOIN transaction AS t ON c.id = t.company_id WH...	16 row(s) returned	0.014 sec / 0.000014...
11:52:46	SELECT country AS pais_veniendo FROM company AS c LEFT JOIN transaction AS t ON c.id = t.company_id WH...	Error Code: 1054. Unknown column 'declined' in 'having clause'	0.0011 sec
11:55:05	SELECT company_name, AVG(amount) AS top_ventas FROM company AS c JOIN transaction AS t ON c.id = t.comp...	1 row(s) returned	0.013 sec / 0.000017...
11:55:28	SELECT company_name, ROUND(AVG(amount), 2) AS top_ventas FROM company AS c JOIN transaction AS t ON c.i...	1 row(s) returned	0.012 sec / 0.000006...

Exercici 3

Utilitzant només subconsultes (sense utilitzar **JOIN**):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```

30 -- S2.N1.03 Utilitzant només subconsultes (sense utilitzar 'JOIN'):
31 -- Mostra totes les transaccions realitzades per empreses d'Alemanya.
32
33 • SELECT *
34 FROM transaction
35 WHERE company_id IN (
36 SELECT id
37 FROM company
38 WHERE country = "Germany") AND declined = 0;
39

```

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
08AFBFD9-28F0-4B29-8BFD-148FB985504D	CcU-3512	b-2222	193	55.5887	-3.37017	2021-01-05 09:38:43	453.04	0
09273F8B-A0F6-486A-8406-200AF84CF533	CcU-3508	b-2222	185	55.0463	-3.03922	2022-06-11 04:38:13	376.87	0
0C69DC28-19F3-4219-B0F9-C22E42743900	CcS-4969	b-2222	388	41.3846	2.16724	2021-09-27 11:13:46	295.87	0
0DF0D4EC-98D0-4B83-A1E9-2668CA9B5A29	CcS-8432	b-2222	1851	41.4178	12.5114	2021-09-10 20:23:28	53.01	0
0F05617D-1A17-444C-AFCD-5B07FF12F68C	CcS-5024	b-2222	443	39.271	-8.46075	2021-04-28 13:26:51	259.49	0
10CEB33C-FF63-4929-B72C-A6629D7D0E09	CcS-7023	b-2222	2442	41.1554	-8.82017	2018-05-02 20:31:17	246.92	0
193CB54E-3E43-42F2-B88A-BE3CD86FBFEB	CcS-5667	b-2222	1086	50.7604	9.96093	2021-03-23 17:05:42	297.19	0
1AD5D338-3BE7-475C-871E-5CCE9998C949	CcS-5035	b-2222	454	52.0968	18.878	2021-10-22 19:11:38	325.49	0
206CC399-6866-4566-AE50-EECED33862946	CcS-6024	b-2222	443	39.0881	-8.12259	2021-04-03 18:10:35	259.49	0
234FA2BA-705B-4A8B-B9A6-6AFA8888CDCF	CcS-4930	b-2222	349	60.1744	18.9563	2020-09-11 23:08:25	653.87	0
277F3800-0872-4E1F-AF63-132C2ACF773E	CcS-4983	b-2222	402	38.7194	-9.14862	2019-05-01 08:05:34	141.86	0
293228F7-84F4-4A6C-9FC6-C6B8B8B8DCD0	CcS-5036	b-2222	455	51.6081	19.16	2022-05-19 04:50:50	678.81	0
29AC348B-9549-406C-A807-F8FFC04EC31F	CcS-8778	b-2222	4197	51.4406	5.46844	2019-12-24 01:11:07	93.44	0
2C3019BD-DA11-4D84-98AE-01BD067F1CFF	CcS-4998	b-2222	417	39.5446	-8.55107	2022-05-06 23:34:34	589.15	0
2E51051F-0A9E-4EA8-98C9-CE776122D04	CcS-5324	b-2222	743	55.444	-3.47504	2019-03-20 19:33:27	13.14	0
35EC568-7D9A-4215-9393-6B1454F13976	CcS-4866	b-2222	285	40.4076	-3.71124	2022-09-10 00:25:41	630.52	0
3583184C-F2EE-4C6B-B1CB-D907D2B9D0E9	CcS-5841	b-2222	1260	59.9892	18.7169	2021-10-28 14:39:35	328.81	0
37AB57F8-F9FF-4FC0-BF9D-43BD0606A471	CcU-3512	b-2222	193	55.0474	-3.41418	2021-01-24 22:21:00	453.04	0

De nuevo volvemos a usar el condicional con el declined=0 ya que nos pide transacciones realizadas, es decir que hayan sido llevadas a cabo. En caso contrario sería simplemente eliminar ese condicional y ya ver en la columna de “declined” si son transacciones rechazadas o no.

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```

69 -- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.
70
71 • SELECT company_id, (SELECT company_name
72 FROM company c
73 WHERE t.company_id = c.id) AS nombre_empresa
74 FROM transaction t
75 WHERE amount > (SELECT AVG(amount)
76 FROM transaction) AND declined = 0
77 GROUP BY company_id;
78

```

company_id	nombre_empresa
b-2230	Fusco Corp.
b-2234	Cornwallis Incorporated
b-2238	Anite Iaculis Nec Foundation
b-2242	Donec Ltd
b-2246	Sed Nunc Ltd
b-2250	Armet Nulla Donec Corporation
b-2254	Natoconet Ridiculus Mus Inc.
b-2258	Vestibulum Lorem PC
b-2262	Gravida Sagittis LLP
b-2266	Mus Aenean Eget Foundation
b-2270	Dis Parturient Institute

En este caso volvemos con el término “transacciones realizadas”.

- Eliminarian del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```

49 -- Eliminarian del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.
50
51 • SELECT id, company_name AS empresa_sin_transacciones
52 FROM company
53 WHERE id NOT IN (SELECT company_id
54 FROM transaction
55 WHERE company_id IS NOT NULL);
56

```

Result Grid

id	empresa_sin_transacciones
company 27	

Action Output

Time	Action	Response
29... 10:35:42	SELECT id, company_name FROM (SELECT company_id FROM transaction) AS empresas_nul WHERE empresas_nul.company_id I...	Error Code: 1054: Unknown column 'id' in 'field list'
29... 10:42:37	SELECT id, company_name FROM company WHERE id NOT IN (SELECT company_id -- revisar la subquery que seguramente no va a...	0 row(s) returned
2951 10:45:38	SELECT COUNT(*) FROM transaction WHERE company_id IS NULL	1 row(s) returned
29... 10:47:51	SELECT id, company_name FROM company WHERE id NOT IN (SELECT company_id FROM transaction WHERE company_id IS NOT...	0 row(s) returned
29... 10:48:28	SELECT id, company_name AS empresa_sin_transacciones FROM company WHERE id NOT IN (SELECT company_id FROM transacti...	0 row(s) returned

Ninguna de las empresas está en el listado para eliminar.

Exercici 4

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

```

92
93 • DROP TABLE IF EXISTS credit_card;
94
95 • CREATE TABLE IF NOT EXISTS credit_card (
96     id VARCHAR(15) PRIMARY KEY,
97     IBAN VARCHAR(35),
98     pan VARCHAR(20),
99     pin VARCHAR(4),
100    cvv VARCHAR(3),
101    expiring_date VARCHAR(10)
102 );
103

```

Añadimos un DROP TABLE IF EXISTS para facilidad en la ejecución del script.

```

57 -- S2.04 La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit.
58 -- La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction"
59 -- i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit".
60 -- Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.
61
62 • CREATE TABLE IF NOT EXISTS credit_card (
63     id VARCHAR(15) PRIMARY KEY,
64     IBAN VARCHAR(35),
65     pan VARCHAR(20),
66     pin VARCHAR(4),
67     cvv VARCHAR(3),
68     expiring_date VARCHAR(10)
69 );

```

Action Output

Time	Action	Response
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9578', 'XX9915395464561105678/0254', '8999808...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9579', 'XX296393091587170202131236', '9690060...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX781258889851950806677358', '5541182...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX915670516405388124398147', '2624305...	1 row(s) affected
79... 11:56:29	SELECT * FROM transactions.credit_card	5000 row(s) returned

Creemos la tabla "credit_card" de forma genérica con VARCHAR y limitamos los caracteres después de haber visto los datos de los distintos campos. Cargamos los datos de forma exitosa.

Local instance 3306 - Warning - not supported

Administration Schemas N1-Ex-1_estructura datos* S2_nivel1_MariaPalmer* N1-Ex-4_datos_introducir_credit* company transaction credit_card

SCHEMAS

Filter objects

- Functions
- hospitales
- sakila
- sys
- transactions
 - Tables
 - company
 - Columns
 - id
 - company_name
 - phone
 - email
 - country
 - website
 - Indexes
 - Foreign Keys
 - Triggers
 - credit_card
 - Columns
 - id
 - IBAN
 - pan
 - pin

Object Info Session

Schema: transactions

1 • SELECT * FROM transactions.credit_card;

Result Grid

id	IBAN	pan	pin	cvv	expiring_date
CcS-4857	XX4857591835292505850771	2314242385113904	1819	467	09/27/25
CcS-4858	XX8581768137002436304025	6582720299715533	3964	817	12/28/28
CcS-4859	XX7826830491423533608370	886184436289642	4983	277	11/25/26
CcS-4860	XX5581950308885324644396	2481155515486458	6876	661	07/27/27
CcS-4861	XX2035182877196191827307	130803031148557	5710	398	04/25/26
CcS-4862	XX4774721462463645409758	6715617009807829	4042	174	11/27/26
CcS-4863	XX1476829694245046207111	3140679819451364	5969	449	12/27/29
CcS-4864	XX6380298983386731196159	5793672133649114	8481	139	02/25/26
CcS-4865	XX7085078596101025280599	5101552687251312	7847	903	11/25/28
CcS-4866	XX4792859188306396406839	8080768801072813	8271	961	03/28/25
CcS-4867	XX603829881631537483717	7791840337661098	4820	862	11/30/25
CcS-4868	XX3929101617300533068044	4160419683151801	6860	549	08/25/25
CcS-4869	XX1569712096111531886465	5892930716233310	4775	448	08/29/25
CcS-4870	XX644363804187732133949	2227240879956178	5872	715	01/25/25
CcS-4871	XX53200528376786065653	694414345556838	7962	643	07/26/26
CcS-4872	XX4081621143155817546334	5035418997919459	3642	191	12/29/26
CcS-4873	XX1362497434153372311591	327032945604823	1038	177	04/25/26
CcS-4874	XX8111642734917968970741	5388631654216752	2445	846	09/30/27

credit_card 1

Action Output

Time	Action	Response
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9578', 'XX296393091587170202131236', '9690060...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9579', 'XX296393091587170202131236', '9690060...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX781258889851950806677358', '5541182...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX915670516405388124398147', '2624305...	1 row(s) affected
79... 11:56:29	SELECT * FROM transactions.credit_card	5000 row(s) returned
79... 11:56:51	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_credit_card FOREIGN KEY (credit_card_id) REFERENCES credit_card...	2572 row(s) affected Records: 2572 Duplicates: 0 Warnings: 0

Query Completed

Introducción de los datos de "dades_introducir_credit" realizada correctamente → 5000 rows de tarjetas de crédito

```

71 • ALTER TABLE transaction
72   ADD CONSTRAINT fk_transaction_credit_card
73   FOREIGN KEY (credit_card_id) REFERENCES credit_card(id);

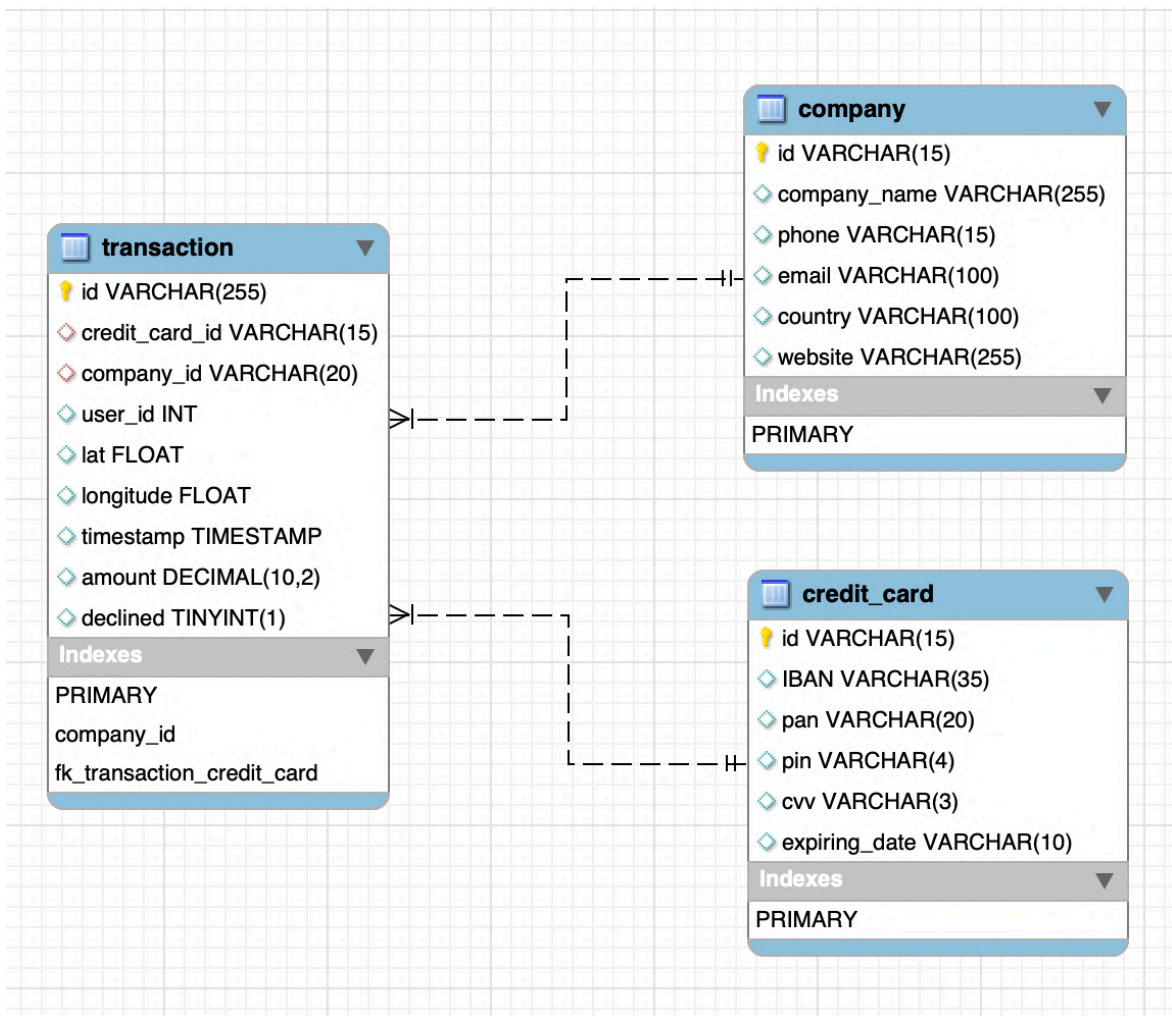
```

100% 59:73

Action Output

Time	Action	Response
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9578', 'XX296393091587170202131236', '9690060...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX781258889851950806677358', '5541182...	1 row(s) affected
79... 11:56:12	INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX915670516405388124398147', '2624305...	1 row(s) affected
79... 11:56:29	SELECT * FROM transactions.credit_card	5000 row(s) returned
79... 11:56:51	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_credit_card FOREIGN KEY (credit_card_id) REFERENCES credit_card...	2572 row(s) affected Records: 2572 Duplicates: 0 Warnings: 0

Modificamos la tabla "transaction" ya que en el ejercicio 1 modificamos la referencia hacia "credit_card" dado que en ese momento no existía la tabla que acabamos de crear. Referenciamos la FK que usaremos para relacionar la tabla "transaction" y "credit_card".



Con la tabla "credit_card" creada y la relación hecha en el paso anterior ya tenemos el diagrama completo con nuestras 3 tablas actuales. En él podemos observar una clara relación de 1:N desde "company" y "credit_card" a "transaction". Esto se debe a que una empresa puede tener múltiples transacciones y lo mismo con una tarjeta de crédito.

Nuestra nueva tabla se compone de distintas columnas, todas ellas con VARCHAR ya que no queremos operar con estos valores y queremos mantenerlos tal y como se introducen. En el caso de "expiring_date" lo usamos también porque el formato de fecha que tenemos en los datos no es como el de DATE.

Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR3234563122135768176999999. Recorda mostrar que el canvi es va realitzar.

user_id 9999

lat 829.999

longitude -117.999

amount 111.11

declined 0

```
85
86 -- 502.06 En la taula "transaction" ingressa una nova transacció amb la següent informació:
87
88 • INSERT INTO company (id, company_name)
89   VALUES ('b-9999', 'Pendiente de alta');
90
91 • INSERT INTO credit_card (id, iban)
92   VALUES ('CcU-9999', 'PENDIENTE');
93
94 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
95   VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0);
96
97 • SELECT *
98   FROM transaction t
99  WHERE t.company_id = 'b-9999';
```

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
108B1D1D-5B23-A76C-55EF-C568E49A99DD	CcU-9999	b-9999	9999	829.999	-117.999	NULL	111.11	0

Time	Action	Response
79... 12:46:57	SELECT * FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned
79... 13:07:56	SELECT * FROM transactions.company WHERE id = 'b-9999'	0 row(s) returned
79... 13:14:11	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0);	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails for database '...', table 'transaction', (company_id): the table 'company' doesn't have a primary key for 'company_id'
79... 13:14:26	INSERT INTO company (id, company_name) VALUES ('b-9999', 'Pendiente de alta')	1 row(s) affected
79... 13:14:45	INSERT INTO credit_card (id, iban) VALUES ('CcU-9999', 'PENDIENTE')	1 row(s) affected
79... 13:14:51	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0);	1 row(s) affected
79... 13:16:15	SELECT * FROM transaction WHERE company_id = 'b-9999'	Error Code: 1054. Unknown column 'company_id' in 'where clause'
79... 13:17:10	SELECT * FROM transaction t WHERE t.company_id = 'b-9999'	1 row(s) returned

Al intentar insertar los datos en la tabla "transaction" nos encontramos con el error 1452 dado que en las tablas padres no existen los id's a los que se tiene que relacionar esta transacción. Entonces insertamos estos id's en sus tablas correspondientes (company y credit_card) y añadimos un comentario en otra columna para dar a entender qué es lo que pasa y que no aparezcan todos los valores de ese id en null. Finalmente comprobamos que los datos precisos se han añadido a la tabla "transaction", dejando todos los campos rellenos menos el de timestamp que no nos lo dan.

Exercici 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordar mostrar el canvi realitzat.

Don't Limit

```

1 • SELECT *
2 • FROM transactions.credit_card;

```

100% 31:2

Result Grid Filter Rows: Search Edit: Export/Import: Fetch rows:

id	IBAN	pan	pin	cvv	expiring_date
CcS-4857	XX4857591835292505850771	2314242385113924	1819	467	09/27/25
CcS-4858	XX8581768137002436094025	6582720299715533	3964	817	12/28/28
CcS-4859	XX7826930491423533609370	8861684536289642	4983	277	11/26/26
CcS-4860	XX5559590398535304645299	2481155515498459	6876	661	07/27/27
CcS-4861	XX2035182877195191627307	1308930301148557	5710	398	04/25/26
CcS-4862	XX4774721462463645409758	671561700807829	4042	174	11/27/26
CcS-4863	XX1476829664245046207111	3140879819451394	5969	449	12/27/29
CcS-4864	XX8380298893385731196159	5793672133649114	8481	139	02/28/26
CcS-4865	XX7085078596101025280599	5101552687251312	7847	903	11/25/28
CcS-4866	XX4792859188206596406839	8080768801072613	9271	961	02/28/25
CcS-4867	XX6038298816319374853717	7761849537661098	4820	862	11/30/25
CcS-4868	XX3929101617300533068044	4160419663151801	6890	549	08/25/25
CcS-4869	XX159671209611531886405	5892930716233310	4775	448	09/29/25
CcS-4870	XX6443683804167732133949	2227240879956178	5872	715	01/25/25
CcS-4871	XX3520209528376786066563	6944143455556838	7992	643	07/26/26
CcS-4872	XX4091621143155817546334	5035418997910459	3642	191	12/29/26

credit_card 31 Apply

Action Output

	Time	Action	Response
✓	79... 13:07:56	SELECT * FROM transactions.company WHERE id = 'b-9999'	0 row(s) returned
✗	79... 13:14:11	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76...	Error Code: 1452. Cannot add or update a child row: a foreign key const...
✓	79... 13:14:26	INSERT INTO company (id, company_name) VALUES ('b-9999', 'Pendiente de alta')	1 row(s) affected
✓	79... 13:14:45	INSERT INTO credit_card (id, iban) VALUES ('CcU-9999', 'PENDIENTE')	1 row(s) affected
✓	79... 13:14:51	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76...	1 row(s) affected
✗	79... 13:16:15	SELECT * FROM transaction WHERE company_id = 'b-9999'	Error Code: 1054. Unknown column 'company.id' in 'where clause'
✓	79... 13:17:10	SELECT * FROM transaction t WHERE t.company_id = 'b-9999'	1 row(s) returned
✓	79... 13:29:18	SELECT * FROM transactions.credit_card	5001 row(s) returned

Tabla "credit_card" antes de ser modificada y donde podemos observar la columna "pan" aún visible.

```

100
101 -- S02.07 Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordra mostrar el canvi realitzat.
102
103 • ALTER TABLE credit_card
104 DROP COLUMN pan;
105
106 • SELECT *
107 FROM credit_card
108 LIMIT 1;

```

100% 9:108

Result Grid Filter Rows: Search Edit: Export/Import:

id	IBAN	pin	cvv	expiring_date
CcS-4857	XX4857591835292505850771	1819	467	09/27/25

credit_card 31 Apply

Action Output

	Time	Action	Response
✓	79... 13:14:26	INSERT INTO company (id, company_name) VALUES ('b-9999', 'Pendiente de alta')	1 row(s) affected
✓	79... 13:14:45	INSERT INTO credit_card (id, iban) VALUES ('CcU-9999', 'PENDIENTE')	1 row(s) affected
✓	79... 13:14:51	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76...	1 row(s) affected
✗	79... 13:16:15	SELECT * FROM transaction WHERE company_id = 'b-9999'	Error Code: 1054. Unknown column 'company.id' in 'where clause'
✓	79... 13:17:10	SELECT * FROM transaction t WHERE t.company_id = 'b-9999'	1 row(s) returned
✓	79... 13:29:18	SELECT * FROM transactions.credit_card	5001 row(s) returned
✓	7971 13:35:54	ALTER TABLE credit_card DROP COLUMN pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓	79... 13:36:45	SELECT * FROM credit_card LIMIT 1	1 row(s) returned

Para deshacernos de la columna "pan" simplemente usamos el ALTER TABLE y el DROP COLUMN. Después mostramos la tabla "credit_card" y observamos que ya no aparece la columna deseada.

Exercici 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

- american_users.csv

- european_users.csv
- companies.csv
- credit_cards.csv
- transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

La taula de products.csv l'utilitzarem més endavant.

1	-- S2.N1.07 Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui,
2	-- almenys 4 taules de les quals puguis realitzar les següents consultes:
3	
4	• CREATE DATABASE IF NOT EXISTS ecommerce;
5	• USE ecommerce;
6	

	Time	Action	Response
1	12:26:28	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__american_u...	Error Code: 1046. No database selected Select the de...
2	12:26:38	USE ecommerce	0 row(s) affected
3	12:26:45	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__american_u...	1010 row(s) affected Records: 1010 Deleted: 0 Skipp...
4	12:26:53	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__european_u...	3990 row(s) affected Records: 3990 Deleted: 0 Skip...
5	12:27:03	SELECT * FROM ecommerce.users_csv	5000 row(s) returned
6	13:04:01	CREATE TABLE IF NOT EXISTS companies_csv (id VARCHAR(255) PRIMARY KEY, company_name VARCHAR(255) NULL, pho...	0 row(s) affected
7	13:04:11	LOAD DATA INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__companies.csv' INT...	Error Code: 1290. The MySQL server is running with t...
8	13:04:22	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__companies...	100 row(s) affected Records: 100 Deleted: 0 Skipp...
9	13:04:26	SELECT * FROM ecommerce.companies_csv	100 row(s) returned
10	13:05:47	CREATE TABLE IF NOT EXISTS credit_cards_csv (id VARCHAR(255) PRIMARY KEY, user_id VARCHAR(255) NULL, iban VARC...	0 row(s) affected
11	13:07:58	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__credit_card...	5000 row(s) affected Records: 5000 Deleted: 0 Skip...
12	13:08:05	SELECT * FROM ecommerce.credit_cards_csv	5000 row(s) returned
13	13:08:18	CREATE TABLE IF NOT EXISTS transactions_csv (id VARCHAR(255) PRIMARY KEY, card_id VARCHAR(255) NULL, business_id...	0 row(s) affected
14	13:08:31	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__transaction...	100000 row(s) affected Records: 100000 Deleted: 0...
15	13:08:39	SELECT * FROM ecommerce.transactions_csv	100000 row(s) returned

Empezamos con un nuevo documento de sql llamado "S2_ecommerce_MariaPalmer" para crear nuestra nueva database "ecommerce" e incluir las nuevas tablas que provienen de los CSV descargados. Como los CSV tratan aspectos de usuarios, transacciones y tarjetas de pago vamos a englobarlo en el nombre de **ecommerce** porque recoge datos de compra en línea.

Indicamos en el documento que solo usaremos esta base de datos "ecommerce" y así nos aseguramos que todas las consultas sean de la misma database.

En total crearemos 4 tablas para esta base de datos con esquema de estrella. Como tenemos 5 documentos CSV vamos a unificar los usuarios americanos y europeos en una sola tabla y dejar el resto como están.

Antes de crear una relación entre las tablas vamos a crearlas de forma genérica para asegurarnos que la carga de los datos a partir de los CSV es correcta. Antes de crear nada en MySQL abrimos los CSV y vemos qué datos y columnas trabajaremos.

```

6
7 CREATE TABLE IF NOT EXISTS users_csv (
8     id VARCHAR(255) PRIMARY KEY,
9     user_name VARCHAR(255) NULL,
10    surname VARCHAR(255) NULL,
11    phone VARCHAR(255) NULL,
12    email VARCHAR (255) NULL,
13    birth_date VARCHAR(255) NULL,
14    country VARCHAR(255) NULL,
15    city VARCHAR(255) NULL,
16    postal_code VARCHAR (15) NULL,
17    address VARCHAR (255) NULL,
18    signup_date VARCHAR(15) NULL,
19    user_segment VARCHAR(25) NULL,
20    income_band VARCHAR(15) NULL
21 );
22
23 SET GLOBAL local_infile = 1;
24
25 LOAD DATA
26 LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADO/1_Especialització/1.3_Sprints/Sprint_2/BBDD/NI-Ex.8__american_users.csv'
27 INTO TABLE users_csv
28 FIELDS TERMINATED BY ','
29 ENCLOSED BY '"'
30 IGNORE 1 ROWS;
31
32 LOAD DATA
33 LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADO/1_Especialització/1.3_Sprints/Sprint_2/BBDD/NI-Ex.8__european_users.csv'
34 INTO TABLE users_csv
35 FIELDS TERMINATED BY ','
36 ENCLOSED BY '"'
37 IGNORE 1 ROWS;
38
39

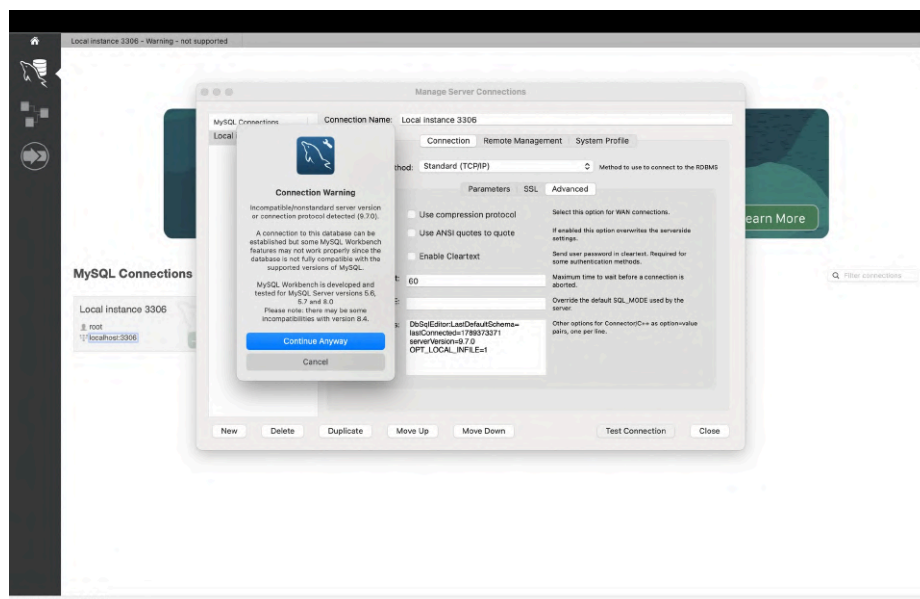
```

	Time	Action	Response	Duration / Fetch Time
1	12:26:28	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADO/1_Especialització/1.3_Sprints/Sprint_2/BBDD/NI-Ex.8__american_u...	Error Code: 1046. No database selected Select the de...	0.00074 sec
2	12:26:38	USE ecommerce	0 row(s) affected	0.00023 sec
3	12:26:45	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADO/1_Especialització/1.3_Sprints/Sprint_2/BBDD/NI-Ex.8__american_u...	1010 row(s) affected Records: 1010 Deleted: 0 Skipp...	0.025 sec
4	12:26:53	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADO/1_Especialització/1.3_Sprints/Sprint_2/BBDD/NI-Ex.8__european_u...	3990 row(s) affected Records: 3990 Deleted: 0 Skip...	0.057 sec
5	12:27:03	SELECT * FROM ecommerce.users_csv	5000 row(s) returned	0.00076 sec / 0.011 s...

Para la primera tabla creamos "users_csv" donde unificaremos los datos de usuarios americanos y europeos. Nos hemos asegurado que los dos documentos tienen la misma estructura y datos en las columnas así que procedemos a crear la tabla con dichos nombres de columna. Únicamente modificamos la columna 2 de "name" a "user_name" para una mayor facilidad en consultas posteriores y legibilidad (a gusto personal).

Vamos a identificarlos con VARCHAR (255) a su mayoría, excepto en algunos casos donde modifiqué el (n) a partir de la observación de los datos.

En cuanto a los datos del CSV he intentado cargarlos primero con un LOAD DATA INFILE pero me daba "Error Code: 1290. The MySQL server is running with the --secure-file-priv option so it cannot execute this statement", por lo que he tenido que cambiarlo a la carga de forma LOCAL. Para ello he añadido el SET GLOBAL y en la configuración de Workbench he permitido la carga local.



Con esta configuración ya he podido cargar los archivos american_users.csv y european_users.csv con el LOAD DATA LOCAL INFILE.

1 • `SELECT * FROM ecommerce.users_csv;`

id	user_name	surname	phone	email	birth_date	country	city	postal_code	address	signup_date	user_segment	income_band
1	Zeus	Gamble	1-262-581-0551	interdum.anim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagittis St.	2014-04-23	young_professional	high
10	Robert	McCarthy	(324) 746-6771	fermentum@protonmail.com	Apr 30, 1984	United States	San Jose	95101	P.O. Box 773	2013-10-19	premium	high
100	Melodie	McLean	1-677-221-7152	rius.varius@google.ca	Sep 15, 1989	United States	San Jose	95101	Ap #644-8492 Sagittis St.	2015-04-20	frequent_shopper	very_high
1000	Ankgrv	Obuinxbp	+48-256-6936	ankgrv.obuinxbp@example.com	May 17, 1970	Germany	Stuttgart	70173	215 Obuinxbp St	2015-10-26	occasional	low
1001	Nlvrfb	Oydaibwg	+94-121-2522	nlvrfb.oydaibwg@example.com	Mar 4, 1994	Germany	Cologne	50667	121 Oydaibwg St	2014-05-22	premium	very_high
1002	ljbfrnd	Jboddzhpv	+70-122-3668	ljbfrnd.jboddzhpv@example.com	Sep 27, 2001	Germany	Munich	80331	412 Jboddzhpv St	2013-02-11	student	medium
1003	Uycig	Sibdyzmj	+58-123-6968	uycig.sibdyzmj@example.com	Jan 20, 1981	Germany	Stuttgart	70173	735 Sibdyzmj St	2014-01-29	family	medium
1004	Yjqrq	Ojzvggj	+77-944-2340	yjqrq.ojzvggj@example.com	Jul 27, 1954	Germany	Munich	80331	685 Ojzvggj St	2014-11-22	established	medium
1005	Mnqktu	Glofegwk	+54-801-2627	mnqktu.glofegwk@example.com	Nov 15, 1962	Portugal	Funchal	9000-001	8 Glofegwk St	2015-01-13	occasional	medium
1006	Rbkocq	Swkmymuj	+38-196-6580	rbkocq.swkmymuj@example.com	Dec 2, 1989	United States	New York	10001	567 Swkmymuj St	2014-12-09	frequent_shopper	medium
1007	Ehcrfp	Xahkzrim	+71-882-6101	ehcrfp.xahkzrim@example.com	Nov 8, 1971	Spain	Sevilla	41001	205 Xahkzrim St	2013-02-16	young_professional	low
1008	Securp	Faohvfy	+76-822-2041	securp.faohvfy@example.com	Nov 13, 1957	Sweden	Stockholm	111 20	535 Faohvfy St	2014-11-28	established	medium
1009	Raznah	Vlqflaqt	+67-294-3155	rsznah.vlqflaqt@example.com	Nov 10, 1952	Spain	Valencia	46001	507 Vlqflaqt St	2013-10-23	established	medium
101	Sarah	Beck	(358) 691-4345	vitae.rius@aol.co.uk	Apr 9, 1983	United States	Philadelphia	19101	665-9047 In	2014-08-17	frequent_shopper	high
1010	Drugzj	Aaesjmca	+78-476-4768	drugzj.aaesjmca@example.com	Dec 23, 1974	United Kingdom	Birmingham	B1 1AA	822 Aaesjmca St	2014-04-29	occasional	medium
1011	Vosctr	Cmnpnfpu	+74-726-6327	vosctr.cmnpnfpu@example.com	Mar 21, 1954	United States	San Jose	95101	775 Cmnpnfpu St	2015-08-14	established	high
1012	Dwiftr	Garypoly	+57-569-7893	dwiftr.garypoly@example.com	Aug 10, 1981	Netherlands	Amsterdam	1011	818 Garypoly St	2014-09-01	young_professional	medium
1013	Kaahoi	Gskltpnu	+56-147-5969	kaahoi.gskltpnu@example.com	Nov 24, 1993	Poland	Wroclaw	50-001	555 Gskltpnu St	2015-03-15	family	medium
1014	Fcxdi	Qqshifey	+69-709-4077	fcxdi.qqshifey@example.com	Mar 6, 1984	France	Nantes	44000	980 Qqshifey St	2013-09-19	young_professional	high
1015	Ydxijl	Yhewlpqn	+84-730-3250	ydxijl.yhewlpqn@example.com	Nov 13, 2000	Canada	Hamilton	L8E 0A1	209 Yhewlpqn St	2013-05-29	young_professional	medium
1016	Ojvthyc	Nppohxim	+83-425-2753	ojvthyc.nppohxim@example.com	Jul 14, 1986	United Kingdom	Liverpool	L1 1AA	125 Nppohxim St	2013-08-07	frequent_shopper	medium
1017	Yadpva	Lgshquc	+83-571-6077	yadpva.lgshquc@example.com	Sep 17, 1981	Italy	Milan	20100	979 Lgshquc St	2015-04-04	frequent_shopper	high
1018	Ugdldo	Zpbkqjgo	+78-537-1497	ugdldo.zpbkqjgo@example.com	Aug 7, 1950	United States	Chicago	60601	47 Zpbkqjgo St	2013-03-08	established	medium
1019	Bpofv	Egshvplk	+43-486-1951	bpofv.egshvplk@example.com	Oct 1, 1981	Spain	Bilbao	48001	556 Egshvplk St	2014-05-01	frequent_shopper	low
102	Jasper	Landry	1-397-795-1118	consectetur.eiusmod@aol.org	Apr 16, 1982	United States	Philadelphia	19101	Ap #374-7325 Sodales...	2014-02-06	young_professional	medium

users_csv 1

Action Output

	Time	Action	Response	Duration / Fetch Time
10	13:05:47	CREATE TABLE IF NOT EXISTS credit_cards_csv (id VARCHAR(255) PRIMARY KEY, user_id VARCHAR(255) NULL, iban VARCH...	0 row(s) affected	0.013 sec
11	13:07:58	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8__credit_card...	5000 row(s) affected Records: 5000 Deleted: 0 Skip...	0.066 sec
12	13:08:05	SELECT * FROM ecommerce.credit_cards_csv	5000 row(s) returned	0.00068 sec / 0.0097...
13	13:08:18	CREATE TABLE IF NOT EXISTS transactions_csv (id VARCHAR(255) PRIMARY KEY, card_id VARCHAR(255) NULL, business_...	0 row(s) affected	0.015 sec
14	13:08:31	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8__transac...	100000 row(s) affected Records: 100000 Deleted: 0...	0.730 sec
15	13:08:39	SELECT * FROM ecommerce.transactions_csv	100000 row(s) returned	0.0012 sec / 0.091 sec
16	13:41:01	SELECT * FROM ecommerce.users_csv	5000 row(s) returned	0.0012 sec / 0.0085 s...

Tabla users_csv con los datos de usuarios americanos y europeos.

39 • `CREATE TABLE IF NOT EXISTS companies_csv (`

40 `id VARCHAR(255) PRIMARY KEY,`

41 `company_name VARCHAR(255) NULL,`

42 `phone VARCHAR(30) NULL,`

43 `email VARCHAR(255) NULL,`

44 `country VARCHAR(50) NULL,`

45 `website VARCHAR(255) NULL,`

46 `merchant_category VARCHAR(30) NULL,`

47 `merchant_price_position VARCHAR(30) NULL`

48 `);`

50

51 • `LOAD DATA`

52 `LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8__companies.csv'`

53 `INTO TABLE companies_csv`

54 `FIELDS TERMINATED BY ','`

55 `ENCLOSED BY ''`

56 `IGNORE 1 ROWS;`

57

Action Output

	Time	Action	Response	Duration / Fetch Time
2	12:26:28	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8__american_u...	Error Code: 1046. No database selected Select the de...	0.00074 sec
3	12:26:38	USE ecommerce	0 row(s) affected	0.00023 sec
4	12:26:45	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8__american_u...	1010 row(s) affected Records: 1010 Deleted: 0 Skip...	0.025 sec
5	12:26:53	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8__european_u...	3990 row(s) affected Records: 3990 Deleted: 0 Skip...	0.057 sec
6	12:27:03	SELECT * FROM ecommerce.users_csv	5000 row(s) returned	0.00076 sec / 0.011 s...
7	13:04:11	CREATE TABLE IF NOT EXISTS companies_csv (id VARCHAR(255) PRIMARY KEY, company_name VARCHAR(255) NULL, pho...	0 row(s) affected	0.024 sec
8	13:04:22	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8__companies...	100 row(s) affected Records: 100 Deleted: 0 Skipp...	0.0031 sec

Para la segunda tabla creamos "companies_csv" con los datos de un único CSV y usamos los mismo nombres de las columnas ya existentes en el documento. De nuevo, usamos de forma general VARCHAR y ajustamos el valor n en algunos de los casos donde ya hemos observado que se puede acotar.

Como en el caso anterior, vamos a cargar el CSV de forma local con la ruta correspondiente.

1 • **SELECT * FROM ecommerce.companies_csv;**

id	company_name	phone	email	country	website	merchant_cat...	merchant_price_posit...
b-2222	Ac Fermentum Incorporated	+64 601 244 6320	donec.porrtitor.tellus@yahoo.net	Germany	https://www.ac-fermentum-incorporated.com	Beauty	mid-market
b-2226	Magna A Neque Industries	+33 823 560 3779	risus.donec.nibh@icloud.org	Australia	https://www.magne-a-neque-industries.com	Travel	budget
b-2230	Fuice Corp.	+29 657 646 4324	rius@protonmail.edu	United States	https://www.fuice-corp.com	Home	mid-market
b-2234	Conwallis In Incorporated	+6 693 982 1255	mauris.ut@aol.co.uk	Germany	https://www.conwallis-in-incorporated.com	Sports	mid-market
b-2238	Ante Iaculis Nec Foundation	+69 914 549 1009	sed.dictum.prim@outlook.ca	New Zealand	https://www.ante-iaculis-nec-foundation.com	Toys	mid-market
b-2242	Donec Ltd	+56 897 341 8418	at.iaculis@hotmail.co.uk	Norway	https://www.donec-ltd.com	Home	mid-market
b-2246	Sed Nunc Ltd	+5 218 900 3193	nibh@yahoo.org	United Kingdom	https://www.sed-nunc-ltd.com	Home	mid-market
b-2250	Amet Nulla Donec Corporation	+96 338 276 8449	mattis.integer.eu@protonmail.net	Italy	https://www.amet-nulla-donec-corporation.com	Gaming	budget
b-2254	Nasoeur Ridiculus Mus Inc.	+63 316 953 7970	suspendisse.dui@icloud.net	United States	https://www.nasoeur-ridiculus-mus-inc.com	Food	premium
b-2258	Vestibulum Lorem PC	+27 788 560 5000	aenean.massa.integer@aol.net	Belgium	https://www.vestibulum-lorem-pc.com	Electronics	premium
b-2262	Gravida Sagittis LLP	+98 204 198 9702	turpis.vitae@google.ca	Sweden	https://www.gravida-sagittis-llp.com	Sports	premium
b-2266	Mus Aenean Eget Foundation	+1 504 922 3000	mi.duis@hotmail.net	Sweden	https://www.mus-aenean-eget-foundation.com	Sports	mid-market
b-2270	Dis Parturient Institute	+69 962 744 6989	purus@protonmail.org	Ireland	https://www.dis-parturient-institute.com	Food	premium
b-2274	Sed LLC	+3 536 122 8140	at@outlook.com	Belgium	https://www.sed-llc.com	Fashion	mid-market
b-2278	Arco LLP	+53 776 813 5178	dui@aol.ca	Norway	https://www.arco-llp.com	Home	mid-market
b-2282	Pretium Neque Corp.	+17 184 177 7500	eielend.nec.malesuada@protonmail.com	Australia	https://www.pretium-neque-corp.com	Electronics	mid-market
b-2286	Fringilla LLC	+1 150 784 1716	gravida@protonmail.co.uk	New Zealand	https://www.fringilla-llc.com	Sports	budget

companies_csv 1

Action Output

	Time	Action	Response	Duration / Fetch Time
4	12:26:53	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_european_u...	3990 row(s) affected Records: 3990 Deleted: 0 Skip...	0.057 sec
5	12:27:03	SELECT * FROM ecommerce.users_csv	5000 row(s) returned	0.00076 sec / 0.011 s...
6	13:04:01	CREATE TABLE IF NOT EXISTS companies_csv (id VARCHAR(255) PRIMARY KEY, company_name VARCHAR(255) NULL, pho...	0 row(s) affected	0.024 sec
7	13:04:11	LOAD DATA INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_companies.csv' INT...	Error Code: 1290. The MySQL server is running with t...	0.0039 sec
8	13:04:22	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_companies...	100 row(s) affected Records: 100 Deleted: 0 Skippe...	0.0031 sec
9	13:04:26	SELECT * FROM ecommerce.companies_csv	100 row(s) returned	0.00069 sec / 0.000...

Tabla "companies_csv" con los datos cargados.

```

57
58 • CREATE TABLE IF NOT EXISTS credit_cards_csv (
59     id VARCHAR(255) PRIMARY KEY,
60     user_id VARCHAR(255) NULL,
61     iban VARCHAR(255) NULL,
62     pan VARCHAR (255) NULL,
63     pin VARCHAR(255) NULL,
64     cvv VARCHAR(255) NULL,
65     track1 VARCHAR (255) NULL,
66     track2 VARCHAR(255) NULL,
67     expiring_date VARCHAR(255) NULL,
68     card_type VARCHAR(255) NULL,
69     card_renewal_flag VARCHAR (255) NULL
70 );
71
72 • LOAD DATA
73 LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_credit_cards.csv'
74 INTO TABLE credit_cards_csv
75 FIELDS TERMINATED BY ','
76 ENCLOSED BY '"'
77 IGNORE 1 ROWS;

```

Action Output

	Time	Action	Response	Duration / Fetch Time
4	12:26:53	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_european_u...	3990 row(s) affected Records: 3990 Deleted: 0 Skip...	0.057 sec
5	12:27:03	SELECT * FROM ecommerce.users_csv	5000 row(s) returned	0.00076 sec / 0.011 s...
6	13:04:01	CREATE TABLE IF NOT EXISTS companies_csv (id VARCHAR(255) PRIMARY KEY, company_name VARCHAR(255) NULL, pho...	0 row(s) affected	0.024 sec
7	13:04:11	LOAD DATA INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_companies.csv' INT...	Error Code: 1290. The MySQL server is running with t...	0.0039 sec
8	13:04:22	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_companies...	100 row(s) affected Records: 100 Deleted: 0 Skippe...	0.0031 sec
9	13:04:26	SELECT * FROM ecommerce.companies_csv	100 row(s) returned	0.00069 sec / 0.000...
10	13:05:47	CREATE TABLE IF NOT EXISTS credit_cards_csv (id VARCHAR(255) PRIMARY KEY, user_id VARCHAR(255) NULL, iban VARC...	0 row(s) affected	0.013 sec
11	13:07:58	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_credit_card...	5000 row(s) affected Records: 5000 Deleted: 0 Skip...	0.066 sec

Para la tercera tabla creamos "credit_cards_csv" con los datos del CSV correspondiente. Las mismas columnas y los mismos nombres. En este caso ya directamente todos en VARCHAR (255) ya que los datos pueden llegar a ser muy variados.

1 • `SELECT * FROM ecommerce.credit_cards_csv;`

id	user_id	iban	pan	pin	cvv	track1	track2	expiring_date	card_type	card_renewal_fl...
CcS-4859	278	SE720084318584364771497282	8861684536289642	4983	277	%B8861684536289642~COFBDGCOFBDG~28...	%B8861684536289642~2502101761665371?	11/26/28	credit	0
CcS-4860	279	ES186574078045314395283427	2481155515498459	6876	661	%B2481155515498459~TUTUTUTUTU~31040...	%B2481155515498459~2602101514414395?	07/27/28	prepaid	0
CcS-4861	280	SE56762210431035196620841	1308930301149557	5710	399	%B1308930301149557~HPOBN2~P0INZ~330...	%B1308930301149557~2605101751305028?	04/26/28	credit	0
CcS-4862	281	NL940396429710439685071325	67156170098078294	4042	174	%B67156170098078294~DMWTDLCMDWT~33...	%B67156170098078294~2210101702370428?	11/26/28	credit	0
CcS-4863	282	FR828211851201116231488674	3140879819451394	5969	449	%B3140879819451394~OXXJDOXXJDO~230...	%B3140879819451394~3210101155648599?	12/29/28	debit	0
CcS-4864	283	SE455021683679182752728201	5793672133649114	8481	139	%B5793672133649114~NHWBVRNHWBYR~30...	%B5793672133649114~2306101806367101?	02/26/28	debit	0
CcS-4865	284	PL151333089612763384460219	5101552687251312	7847	903	%B5101552687251312~MJDHJHJHJH~33...	%B5101552687251312~3010101664867107?	11/26/28	debit	0
CcS-4866	285	ES185905913522238333641620	8080768801072613	9271	961	%B8080768801072613~MEBGOZMEBGOZ~32...	%B8080768801072613~2610101715885695?	02/25/28	debit	1
CcS-4867	286	NL7600067565349226474321	7761849537661098	4320	862	%B7761849537661098~LDRSOLLDRSOLL~281...	%B7761849537661098~2206101710411371?	11/25/28	credit	1
CcS-4868	287	DE60402393098318335747030	4160419683151801	6860	549	%B4160419683151801~KQJCKJCKJCK~270...	%B4160419683151801~2508101278587927?	08/25/28	credit	1
CcS-4869	288	IT280405825351365860527091	5892930716233310	4775	448	%B5892930716233310~DNSCAVDNSCAV~250...	%B5892930716233310~2403101253758128?	09/25/28	premium...	1
CcS-4870	289	DE312678675102861012587987	2227240879956178	5872	715	%B2227240879956178~NNZJTRNNZJTR~340...	%B2227240879956178~3411101430032222?	01/25/28	debit	1
CcS-4871	290	NL841882812537051501857688	694414345556838	7992	643	%B694414345556838~FRFEEUFRFEEU~270...	%B694414345556838~3212101363716238?	07/26/28	credit	0
CcS-4872	291	PT894639039284374217174740	5035418997919459	3642	191	%B5035418997919459~SXVXZSXVXZ~33080...	%B5035418997919459~2812101790314055?	12/26/28	credit	0
CcS-4873	292	ES561527429673850446657189	3270329845604923	1038	177	%B3270329845604923~XQTOZDXQTOZ~280...	%B3270329845604923~3112101515908414?	04/26/28	credit	0
CcS-4874	293	IT939133562902807152613985	5388631654216752	2445	846	%B5388631654216752~UTUPBJUTUPBJ~340...	%B5388631654216752~3411101437145559?	09/27/28	credit	0
CcS-4875	294	FR146381647733726778398375	9466568875132901	2205	142	%B9466568875132901~FKXWAHFKXWAH~29...	%B9466568875132901~2210101523538851?	08/28/28	debit	0

credit_cards_csv 1

Action Output

	Time	Action	Response	Duration / Fetch Time
5	12:27:03	SELECT * FROM ecommerce.users_csv	5000 row(s) returned	0.00076 sec / 0.011 s...
6	13:04:01	CREATE TABLE IF NOT EXISTS companies_csv (id VARCHAR(255) PRIMARY KEY, company_name VARCHAR(255) NULL, pho...	0 row(s) affected	0.024 sec
7	13:04:11	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_companies.csv' INT...	Error Code: 1290. The MySQL server is running with t...	0.0039 sec
8	13:04:22	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_companies...	100 row(s) affected Records: 100 Deleted: 0 Skipp...	0.0031 sec
9	13:04:26	SELECT * FROM ecommerce.companies_csv	100 row(s) returned	0.00069 sec / 0.000...
10	13:05:47	CREATE TABLE IF NOT EXISTS credit_cards_csv (id VARCHAR(255) PRIMARY KEY, user_id VARCHAR(255) NULL, iban VARC...	0 row(s) affected	0.013 sec
11	13:07:58	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_credit_card...	5000 row(s) affected Records: 5000 Deleted: 0 Skipp...	0.066 sec
12	13:08:05	SELECT * FROM ecommerce.credit_cards_csv	5000 row(s) returned	0.00068 sec / 0.0097...

Tabla "credit_cards_csv".

79 • `CREATE TABLE IF NOT EXISTS transactions_csv (`

80 `id VARCHAR(255) PRIMARY KEY,`

81 `card_id VARCHAR(255) NULL,`

82 `business_id VARCHAR(255) NULL,`

83 `timestamp VARCHAR(255) NULL,`

84 `amount VARCHAR(255) NULL,`

85 `declined VARCHAR(255) NULL,`

86 `product_ids VARCHAR(255) NULL,`

87 `user_id VARCHAR(255) NULL,`

88 `lat VARCHAR(255) NULL,`

89 `longitude VARCHAR(255) NULL,`

90 `discount_amount VARCHAR(255) NULL,`

91 `tax_amount VARCHAR(255) NULL,`

92 `shipping_amount VARCHAR(255) NULL,`

93 `channel VARCHAR(255) NULL,`

94 `campaign_id VARCHAR(255) NULL,`

95 `device_type VARCHAR(255) NULL,`

96 `is_international VARCHAR(255) NULL,`

97 `decline_reason VARCHAR(255) NULL,`

98 `distance_km VARCHAR(255) NULL`

99 `);`

100

101 • `LOAD DATA`

102 `LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_transactions.csv'`

103 `INTO TABLE transactions_csv`

104 `FIELDS TERMINATED BY ',';`

105 `ENCLOSED BY ''`

106 `IGNORE 1 ROWS;`

107

Action Output

	Time	Action	Response	Duration / Fetch Time
11	13:07:58	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_credit_card...	5000 row(s) affected Records: 5000 Deleted: 0 Skip...	0.066 sec
12	13:08:05	SELECT * FROM ecommerce.credit_cards_csv	5000 row(s) returned	0.00068 sec / 0.0097...
13	13:08:18	CREATE TABLE IF NOT EXISTS transactions_csv (id VARCHAR(255) PRIMARY KEY, card_id VARCHAR(255) NULL, business_id...	0 row(s) affected	0.015 sec
14	13:08:31	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex-8_transaction...	100000 row(s) affected Records: 100000 Deleted: 0...	0.730 sec

Por último creamos la última tabla "transactions_csv" con los datos del CSV correspondiente. De nuevo asignamos las mismas columnas con VARCHAR (255) de forma general.

La diferencia en este caso es en la carga de los datos. En el documento transactions no se usa ',' para separar los campos, si no que se usa ';'. Lo modificaremos para que los datos se carguen en el campo que le corresponde y evitar errores o confusiones.

N1-Ex.8_transactions.csv

```
id;card_id;business_id;timestamp;amount;declined;product_ids;user_id;lat;longitude;discount_amount;tax_amount;shipping_amount;channel;campaign_id;device_type;is_international;decline_reason;distance_km
CDDA7E40-544D-47BB-A4ED-671DD8A950D9;CcS-6894;b-2466;2018-12-12 08:05:17;176.45;0;75, 73, 98;2313;59.62050974356148;16.559977155728436;20.72;35.29;0.00;store;CHRISTMAS_PEAK;pos;0;;129.2509456357-8E9B-475A-8257-87A023699964;CcS-5135;b-2342;2024-05-20 22:49:08;206.65;0;3;554;45.76458841901318;4.843056518287656;3.08;33.61;4.99;online;NO_CAMPAIGN;moble;0;;0.57
```

N1-Ex.8_credit_cards.csv

```
id,user_id,iban,pan,pin,cvv,track1,track2,expiring_date,card_type,card_renewal_flag
CcU-2938,275,CA137550525951024363769804,5424465566813633,3257,984,%B8383712448554646^WovsxejDpwi
ey^8604114277,%B7653863056044187=800716333673,04/27/28,credit,1
CcU-2945,274,CA657195557325315948524978,5142423821948828,9080,887,%B4621311609958661^UftuyfsSeim
xn^061062824177,%B4149568437843501=510714033071,01/27/28,premium_credit,1
CcU-2952,273,CA749863982219203741201012,4556 453 55
5287,4598,438,%B2183285104307501^CdddytcUxwfdq^590795543079,%B6778580257827162=69068597400?
7,08/29/28,credit,1
CcU-2959,272,CA164763652898476863375896,372461377349375,3583,667,%B7281111956795320^XocddijBckec
d^0901625373,%B4246154489281853=280522391678,08/29/28,credit,1
CcU-2966,271,CA913060993220307056913286,448566 886747
7265,4900,130,%B4728932322756223^JhlgvsuFbmwgj^7202289494377,%B2318571115599881=8908215784?
5,04/26/28,credit,1
CcU-2973,270,CA365825223870062682219085,544 58654 54343
```

Diferencia de la separación de campos de transactions (;) comparado con por ejemplo credit_cards (,).

1 • SELECT * FROM ecommerce.transactions_csv;

id	card_id	business_id	timestamp	amount	declined	product_ids	user_id	lat	longitude	discount_amo...	tax_amount	shipping_amo...
00043449-2949-494B-A5DD-ASBAE3BB19DD	CcS-9294	b-2458	2024-08-28 07:16:46	433.51	0	16, 28, 97, 87	4713	46.19992926158272	1.4355402821327807	34.17	72.25	0.00
000447FE-B850-4D0F-85DE-C7ED0EE1CAAD	CcS-5019	b-2370	2015-12-21 20:07:18	172.56	0	66, 69, 87	438	41.59720554463741	12.22175994259365	19.92	29.86	6.99
00045DEB-ED2E-4F2F-8186-CE07D4D875D0	CcS-6609	b-2390	2020-07-14 15:37:45	332.38	0	30, 11, 16, 81	2118	29.757295899964348	-95.37963676492454	16.10	22.47	0.00
000481C3-1C26-4FEF-83A0-4CDD0EB004BD	CcS-6696	b-2230	2017-09-04 19:44:53	175.00	0	72	2115	53.54888376797025	-113.50305274646564	1.05	14.45	0.00
00051AA4-9CBE-4268-B070-C38062A193E2	CcS-7606	b-2266	2017-01-05 18:19:25	158.55	0	18	3025	52.20836951854172	5.690806474241335	25.31	25.96	8.99
0008A312-EDFE-4A4F-BC99-E9392CC3CA4D	CcU-3358	b-2598	2023-05-23 04:51:43	317.97	0	35, 33, 19	215	53.553485560258014	-113.4991339874781	2.87	26.25	0.00
0009A151-9BCF-4E31-9053-A86FF77FAAB	CcS-7509	b-2546	2023-12-31 00:06:36	457.51	0	93, 55, 28, 91	2928	52.522483304690255	13.552942314307499	5.52	79.40	0.00
0009D494-6245-4DF9-955D-2C0A4191CFFB	CcS-6483	b-2526	2017-07-18 07:52:02	225.49	0	55, 8, 72	3902	45.49200875032752	-73.57083686984957	2.85	17.55	12.99
000A1DEC-CD86-4AB2-A619-71DAB8DA282	CcS-6467	b-2558	2018-09-08 05:29:58	374.41	0	46, 56, 73	1886	55.74247746345254	-3.30093680889247	27.93	62.40	0.00
000A1E64-1414-40B0-9D92-5678A4D958E2	CcS-5966	b-2550	2022-09-17 04:02:19	378.00	0	6, 89, 19	1385	52.08214987185822	5.284238674948618	57.31	65.60	0.00
000A5879-3472-41D9-AF60-42D3503B543C	CcU-4569	b-2590	2020-02-07 23:03:45	185.46	0	58, 61, 55	42	39.94902273793568	-75.17193552521194	1.62	11.66	12.99
000AE0D4-F06E-4146-904A-5A2FC73110D7	CcS-8134	b-2426	2017-07-10 07:11:41	187.69	0	62	3503	49.37477928882258	-123.26964080097178	20.41	15.17	3.99
000BCB9F-AAAB-4E8E-A148-2E7AD4E1E385	CcS-6020	b-2598	2019-05-28 04:23:02	108.70	0	36	1439	33.44717421182664	-112.07205636490832	1.39	6.88	6.99
000BD852-E258-4785-93D9-27081425D458	CcS-5445	b-2330	2024-07-31 09:45:39	117.35	0	8, 91	864	51.43172538140102	5.474892557826071	0.92	19.15	6.99
000DAABF-ABEA-4F3D-A090-B9C0705D9E9	CcS-6404	b-2454	2023-07-26 22:39:50	345.24	0	76, 19, 89	1823	41.73921421715434	12.629293242939156	15.20	62.26	0.00
000E44B7-43B4-443D-8B82-ABAD403D081	CcS-7604	b-2458	2018-08-01 11:03:20	194.89	0	3	3023	59.507385487381185	18.032571684402404	8.72	32.48	0.00
000E64A2-2C13-4306-BB25-B292ED79413D	CcS-6737	b-2386	2018-05-30 16:09:08	119.75	0	54, 17	2156	45.49919845970472	-73.57193491704714	1.58	9.31	6.99
000E65B9-4747-42C4-20C2-17A01FC37CF	CcS-6096	b-2502	2019-11-28 07:13:00	243.45	0	88, 4	1515	39.40736072785744	-8.463495095541386	41.11	45.52	0.00
0010CCE6-CF15-4F25-85CE-A1530BEF4D6B	CcS-8139	b-2438	2021-10-03 15:58:34	318.67	0	52, 3, 91, 75	3558	51.1838461739755	10.508633919547709	4.08	50.88	0.00
0011D0BC-79A-4DEB-B13E-C8CC0892CFC	CcU-3449	b-2606	2020-05-30 23:27:14	217.02	0	64	202	53.55509896048705	-113.48687695160032	2.93	17.34	6.99
0011D40F-1F79-4802-670A-43320FE370FF	CcS-8725	b-2366	2023-05-13 20:30:16	77.96	0	85	4144	59.608469883751095	16.555130397121148	0.25	15.59	0.00
00126FB9-71D0-4EBC-AA92-00DAD11F1C40	CcS-7057	b-2502	2018-03-13 02:51:18	123.67	0	71	2476	51.49648419187042	18.86307383455276	6.43	22.38	3.99
00130606-1ADD-46D0-85E6-00ADAD11F1C3A	CcU-3631	b-2366	2021-11-23 01:07:15	318.24	0	80, 74, 92	176	55.169512113128064	-3.691287787721104	75.68	53.04	0.00
00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	2020-03-25 10:43:43	512.67	0	74, 34, 13	318	41.37810524421494	12.44897617243859	6.14	92.45	0.00

transactions_csv 1

Action Output

	Time	Action	Response	Duration / Fetch Time
11	13:07:58	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especializació/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8_credit_card...	5000 row(s) affected Records: 5000 Deleted: 0 Skip...	0.066 sec
12	13:08:05	SELECT * FROM ecommerce.credit_cards_csv	5000 row(s) returned	0.00068 sec / 0.0097...
13	13:08:18	CREATE TABLE IF NOT EXISTS transactions_csv (id VARCHAR(255) PRIMARY KEY, card_id VARCHAR(255) NULL, business_id...	0 row(s) affected	0.015 sec
14	13:08:31	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especializació/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8_transactions...	100000 row(s) affected Records: 100000 Deleted: 0...	0.730 sec
15	13:08:39	SELECT * FROM ecommerce.transactions_csv	100000 row(s) returned	0.0012 sec / 0.091 sec

Tabla "transactions_csv" con los datos correspondientes en cada campo.

```

108 -- ¿Transacciones con tarjeta que no existe en credit_cards_csv?
109 • SELECT COUNT(*)
110 FROM transactions_csv t
111 LEFT JOIN credit_cards_csv cc
112 ON t.card_id = cc.id
113 WHERE cc.id IS NULL;
114
115 -- ¿Transacciones con tarjeta que no existe en companies_csv?
116 • SELECT COUNT(*)
117 FROM transactions_csv t
118 LEFT JOIN companies_csv c
119 ON t.business_id = c.id
120 WHERE c.id IS NULL;
121
122 -- ¿Transacciones con tarjeta que no existe en users_csv?
123 • SELECT COUNT(*)
124 FROM transactions_csv t
125 LEFT JOIN users_csv u
126 ON t.user_id = u.id
127 WHERE u.id IS NULL;
128

```

Result Grid

Count(*)
0

Result 3

Action Output

	Time	Action	Response	Duration / Fetch Time
1	10:25:30	SELECT COUNT(*) FROM transactions_csv t LEFT JOIN credit_cards_csv c ON t.card_id = c.id WHERE c.id IS NULL	1 row(s) returned	0.306 sec / 0.00000...
2	10:26:48	SELECT * FROM ecommerce.companies_csv	100 row(s) returned	0.00096 sec / 0.000...
3	10:26:56	SELECT * FROM ecommerce.users_csv	5000 row(s) returned	0.00074 sec / 0.0069...
4	10:27:10	SELECT * FROM ecommerce.transactions_csv	100000 row(s) returned	0.00049 sec / 0.0096...
5	10:29:57	SELECT COUNT(*) FROM transactions_csv t LEFT JOIN companies_csv c ON t.business_id = c.id WHERE c.id IS NULL	1 row(s) returned	0.095 sec / 0.00000...
6	10:30:57	SELECT COUNT(*) FROM transactions_csv t LEFT JOIN users_csv u ON t.user_id = c.id WHERE u.id IS NULL	Error Code: 1054. Unknown column 'c.id' in 'on clause'	0.0015 sec
7	10:31:15	SELECT COUNT(*) FROM transactions_csv t LEFT JOIN users_csv u ON t.user_id = u.id WHERE u.id IS NULL	1 row(s) returned	0.119 sec / 0.000011...

Comprobamos que no tenemos datos huérfanos antes de hacer las relaciones entre las tablas dimensiones y la de "transactions_csv" que va ser el centro de la estrella.

```

136
137 • ALTER TABLE transactions_csv
138 ADD CONSTRAINT fk_transactions_users
139 FOREIGN KEY (user_id) REFERENCES users_csv(id);

```

Action Output

	Time	Action	Response	Duration / Fetch Time
5	10:29:57	SELECT COUNT(*) FROM transactions_csv t LEFT JOIN companies_csv c ON t.business_id = c.id WHERE c.id IS NULL	1 row(s) returned	0.095 sec / 0.00000...
6	10:30:57	SELECT COUNT(*) FROM transactions_csv t LEFT JOIN users_csv u ON t.user_id = c.id WHERE u.id IS NULL	Error Code: 1054. Unknown column 'c.id' in 'on clause'	0.0015 sec
7	10:31:15	SELECT COUNT(*) FROM transactions_csv t LEFT JOIN users_csv u ON t.user_id = u.id WHERE u.id IS NULL	1 row(s) returned	0.119 sec / 0.000011...
8	10:43:22	ALTER TABLE transactions_csv ADD CONSTRAINT fk_transactions_credit_cards FOREIGN KEY (card_id) REFERENCES credit_car...	100000 row(s) affected Records: 100000 Duplicates:...	1.240 sec
9	10:43:34	ALTER TABLE transactions_csv ADD CONSTRAINT fk_transactions_companies FOREIGN KEY (business_id) REFERENCES compan...	100000 row(s) affected Records: 100000 Duplicates:...	2.819 sec
10	10:43:42	ALTER TABLE transactions_csv ADD CONSTRAINT fk_transactions_users FOREIGN KEY (user_id) REFERENCES users_csv(id)	100000 row(s) affected Records: 100000 Duplicates:...	4.635 sec

Alteramos la tabla central "transactions_csv" para hacer la relación N:1 hacia las otras 3 tablas de nuestra base de datos con la foreign key correspondiente.



Finalmente conseguimos nuestra base de datos **ecommerce** relacionada en un modelo de estrella, donde "transactions_csv" es nuestra tabla de hechos y se relaciona a las tablas dimensiones N:1 a través de los "id" de cada una de ellas. Podemos observar esos "id" de las otras 3 tablas como claves foraneas.

Nota: ¿Se podría haber hecho la creación de las tablas directamente con las FK ya incluidas desde un principio? Es muy probable. Para futuros ejercicios ya lo haré así directamente pero quería comprobar que los datos se cargaran bien antes de tener las relaciones entre tablas.

Exercici 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

142	
143	SELECT user_id, (SELECT user_name
144	FROM users_csv u
145	WHERE t.user_id = u.id) AS name, COUNT(id) AS total_transactions
146	FROM transactions_csv t
147	WHERE declined = 0
148	GROUP BY 1
149	HAVING COUNT(id) > 80;
150	
Result Grid	
user_id	name
185	Molly
289	Dwight
318	Bryn
454	Stefan
Result 13	
Action Output	
Time	Action
20 11:21:43	SELECT user_id, (SELECT user_name, surname FROM users_csv u WHERE t.user_id = u.id) AS top_users FROM transactions_csv t
21 11:23:55	SELECT user_id, (SELECT user_name, surname FROM users_csv u WHERE t.user_id = u.id) AS top_users FROM transactions_csv t
22 11:29:46	SELECT id, user_name, surname, COUNT(t.user_id) AS user_transactions FROM users_csv u JOIN transactions_csv t ON u.id = t.user_id
23 11:30:03	SELECT u.id, u.user_name, u.surname, COUNT(t.user_id) AS user_transactions FROM users_csv u JOIN transactions_csv t ON u.id = t.user_id
24 11:30:31	SELECT u.id, u.user_name, u.surname, COUNT(t.user_id) AS transactions FROM users_csv u JOIN transactions_csv t ON u.id = t.user_id
25 11:31:01	SELECT u.id, u.user_name, u.surname, COUNT(t.user_id) FROM users_csv u JOIN transactions_csv t ON u.id = t.user_id WHERE CO...
26 11:33:56	SELECT u.id, u.user_name, u.surname, COUNT(t.id) FROM users_csv u LEFT JOIN transactions_csv t ON u.id = t.user_id WHERE CO...
27 11:34:10	SELECT u.id, u.user_name, u.surname, COUNT(t.id) FROM users_csv u LEFT JOIN transactions_csv t ON u.id = t.user_id WHERE CO...
28 11:34:19	SELECT u.id, u.user_name, u.surname, COUNT(t.id) FROM users_csv u LEFT JOIN transactions_csv t ON u.id = t.user_id WHERE CO...
29 11:34:21	SELECT u.id, u.user_name, u.surname, COUNT(t.id) FROM users_csv u LEFT JOIN transactions_csv t ON u.id = t.user_id WHERE CO...
30 11:34:39	SELECT u.id, u.user_name, u.surname, COUNT(t.id) FROM users_csv u LEFT JOIN transactions_csv t ON u.id = t.user_id WHERE CO...
31 11:43:06	SELECT id, user_name, surname, (SELECT id, COUNT(user_id) FROM transactions_csv t WHERE t.user_id = ...)
32 11:45:04	SELECT t.id, COUNT(user_id) AS count FROM transactions_csv t WHERE count > 80 GROUP BY 1
33 11:45:20	SELECT t.id, COUNT(user_id) AS "count" FROM transactions_csv t WHERE "count" > 80 GROUP BY 1
34 11:45:54	SELECT user_id, COUNT(id) AS "count" FROM transactions_csv t WHERE "count" > 80 GROUP BY 1
35 11:59:15	SELECT user_id, COUNT(id) FROM transactions_csv t GROUP BY 1 HAVING COUNT(id) > 80
36 12:04:19	SELECT user_id, COUNT(id) FROM transactions_csv t WHERE declined = 0 GROUP BY 1 HAVING COUNT(id) > 80
37 12:05:49	SELECT user_name, surname, (SELECT user_id, COUNT(id) FROM transactions_csv t WHERE declined = 0 GROUP BY 1 HAVING CO...
38 12:10:53	SELECT user_id, COUNT(id), (SELECT user_name FROM users_csv u WHERE t.user_id = u.id) AS name FROM transactions_csv t
39 12:13:49	SELECT user_id, COUNT(id) AS total_transactions, (SELECT user_name FROM users_csv u WHERE t.user_id = u.id)...
40 12:14:24	SELECT user_id, (SELECT user_name FROM users_csv u WHERE t.user_id = u.id) AS name, COUNT(id) AS total_transactions F...

Después de darme de cabeza con una subconsulta aparentemente simple (queriendo usar el WHERE por narices y olvidando por completo que existe el HAVING...) por fin conseguimos nuestro listado de usuarios con más de 80 transacciones efectivas.

El enunciado no especifica de forma explícita si se refiere a transacciones totales o solo las que han sido efectuadas, así que he añadido el condicional de que no hayan sido “declined”. En todo caso, usando o no ese condicional el resultado es el mismo (Action Output num.35 y num.36).

Para esta consulta se han usado 2 tablas (transactions_csv y users_csv) para obtener el número total de transacciones y el nombre del usuario que las ha realizado.

Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```

150
151 -- S2.N1.10 Mostra la mitjana d'amount per IBAN de les targetes de cr dit a la companyia Donec Ltd, utilitza almenys 2 taules.
152 • SELECT iban, ROUND(AVG(amount), 2) AS avg_amount, company_name
153 FROM transactions_csv t
154 JOIN companies_csv c
155 ON t.business_id = c.id
156 JOIN credit_cards_csv cc
157 ON t.card_id = cc.id
158 WHERE c.company_name = 'Donec Ltd' AND declined = 0
159 GROUP BY 1;

```

iban	avg_amount	company_name
CA901355773596771626108990	361.01	Donec Ltd
US729212642816955651428791	142.56	Donec Ltd
US989450932309332768119251	246.9	Donec Ltd
US581097432698325541142297	154.01	Donec Ltd
CA441226287105857473570574	250.51	Donec Ltd
CA332171982352073015086775	203.95	Donec Ltd
US985359628701198880628720	438.16	Donec Ltd
US78811972997693675524914	520.37	Donec Ltd
CA967886529258577442049868	204.63	Donec Ltd
US483990742400388023162032	165.08	Donec Ltd
US941589234795722657349646	253.93	Donec Ltd
CA707350503853933968563487	290.57	Donec Ltd
US225174296147002265631190	240.32	Donec Ltd
CA818325250352269640720088	86.52	Donec Ltd
US570082429286901600591866	288.19	Donec Ltd
US183391626605377230916689	83.54	Donec Ltd

Time	Action	Response	Duration / Fetch Time
01:26:29	SELECT pr_id,	100 row(s) returned	0.173 sec / 0.000044...
01:29:47	SELECT pr_id,	100 row(s) returned	0.172 sec / 0.000044...
01:31:22	SELECT pr_id,	100 row(s) returned	0.173 sec / 0.000042...
11:45:34	SELECT iban, ROUND(AVG(amount), 2) AS avg_amount, company_name FROM transactions_csv t JOIN companies_csv c ON t.business_id = c.id JOIN credit_cards_csv cc...	367 row(s) returned	0.044 sec / 0.00064...

Para esta consulta utilizamos 3 tablas a partir de JOIN para conseguir la media de amount para cada uno de los IBAN de la compa  a mencionada en el enunciado. Para ello usaremos el condicional donde limitamos los resultado solo a la empresa llamada 'Donec Ltd' y agrupamos por el "iban".

▼ Nivell 2 ★★

Exercici 1

Identifica els cinc dies que es va generar la quantitat m s gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacci  juntament amb el total de les vendes.

```

305 -- SPRINT 2 - NIVELL 2
306 -- S2.N2.01 Identifica els cinc dies que es va generar la quantitat m s gran d'ingressos a l'empresa per vendes.
307 -- Mostra la data de cada transacci  juntament amb el total de les vendes.
308
309 • SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS top_amount
310 FROM transactions_csv
311 WHERE declined = 0
312 GROUP BY 1
313 ORDER BY 2 DESC
314 LIMIT 5;

```

fecha	top_amount
2023-03-20	15728.4
2019-11-18	15447.88
2022-12-13	14944
2019-03-18	14896.47
2017-12-20	14367.96

Time	Action	Response	Duration / Fetch Time
13:16:47	SELECT (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa, amount, (SELECT ROUND(A...	1222 row(s) returned	0.0078 sec / 0.0029...
13:19:55	SELECT (SELECT DISTINCT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa, amount, (SELECT...	1222 row(s) returned	0.0071 sec / 0.0047 s...
13:20:12	SELECT distinct (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa, amount, (SELECT RO...	1214 row(s) returned	0.011 sec / 0.0017 sec
00:01:09	SELECT company_id, (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa FROM tr...	1222 row(s) returned	0.030 sec / 0.0016 sec
00:01:42	SELECT company_id, (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa FROM tr...	100 row(s) returned	0.017 sec / 0.000021...
00:03:33	SELECT company_id, (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa FROM tr...	100 row(s) returned	0.014 sec / 0.000024...
00:23:33	SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS top_amount FROM transactions_csv WHERE declined = 0 GROUP...	5 row(s) returned	0.084 sec / 0.000011...

Para esta consulta necesitamos el "timestamp" de la tabla "transactions_csv" sin la informaci n horaria as  que lo transformamos a fecha con DATE() y as  conseguimos agrupar por d as tal y

como pide el enunciado.

Exercici 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
316 -- S2.N2.02 Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb
317 -- un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024.
318 -- Ordena els resultats de major a menor quantitat.
319
320 • SELECT company_name, phone, country, DATE(timestamp) AS date, amount
321 FROM companies_csv c
322 JOIN transactions_csv t
323 ON c.id = t.business_id
324 WHERE amount BETWEEN 350 AND 400 AND declined = 0
325 AND DATE(timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
326 ORDER BY amount DESC;
327
```

100% 16:313

Result Grid Filter Rows: Search Export:

company_name	phone	country	date	amount
Fringilla LLC	+1 150 784 1716	New Zealand	2015-04-29	390.33
Auctor Mauris Vel LLP	+32 133 447 8238	United States	2024-03-13	383.58
Mauris Institute	+75 473 619 9472	Sweden	2015-04-29	369.20
Pede Cum Ltd	+26 625 798 6126	Norway	2018-07-20	358.84
Notus El Maeguada Ltd	+52 818 402 1638	Netherlands	2024-03-13	352.21
Yompor Diam Institute	+89 848 141 3138	Netherlands	2024-03-13	350.71

Result 8 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
1	13:16:47	SELECT (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa, amount, (SELECT ROUND(A...	1222 row(s) returned	0.0078 sec / 0.0029...
2	13:19:55	SELECT (SELECT DISTINCT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa, amount, (SELECT...	1222 row(s) returned	0.0071 sec / 0.0047 s...
3	13:20:12	SELECT distinct (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa, amount, (SELECT RO...	1214 row(s) returned	0.011 sec / 0.0017 sec
4	00:01:09	SELECT company_id, (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa FROM tr...	1222 row(s) returned	0.030 sec / 0.0016 sec
5	00:01:42	SELECT company_id, (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa FROM tr...	100 row(s) returned	0.017 sec / 0.000021...
6	00:03:33	SELECT company_id, (SELECT company_name FROM company c WHERE t.company_id = c.id) AS nombre_empresa FROM tr...	100 row(s) returned	0.014 sec / 0.000024...
7	00:23:33	SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS top_amount FROM transactions_csv WHERE declined = 0 GROUP...	5 row(s) returned	0.064 sec / 0.000011...
8	00:25:12	SELECT company_name, phone, country, DATE(timestamp) AS date, amount FROM companies_csv c JOIN transactions_csv t ON c.L...	6 row(s) returned	0.048 sec / 0.000003...

En este caso volvemos a necesitar los datos temporales únicamente con fecha para poder usarlo en el condicional juntamente con el de "amount". Mostramos los resultados de mayor a menor cantidad con el orden descendiente.

```
160
161 -- S2.N2.02 Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb
162 -- un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024.
163 -- Ordena els resultats de major a menor quantitat.
164
165 • SELECT company_name, phone, country, DATE(timestamp) AS fechas, amount
166 FROM companies_csv c
167 JOIN transactions_csv t
168 ON c.id = t.business_id
169 WHERE amount BETWEEN 350 AND 400
170 AND DATE(timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
171 ORDER BY amount DESC;
172
173
```

company_name	phone	country	fechas	amount
Fringilla LLC	+1 150 784 1716	New Zealand	2015-04-29	390.33
Auctor Mauris Vel LLP	+32 133 447 6236	United States	2024-03-13	383.58
Mauris Institute	+75 473 819 9472	Sweden	2015-04-29	369.20
Pede Cum Ltd	+26 625 798 6126	Norway	2018-07-20	358.64
Nelus Et Maesuada Ltd	+52 818 402 1638	Netherlands	2024-03-13	352.21
Tempor Diam Institute	+89 848 141 3136	Netherlands	2024-03-13	350.71

Time	Action	Response	Duration / Fetch Time
11:03:03	SELECT company_name, phone, country, DATE(timestamp) AS fechas, amount FROM companies_csv c LEFT JOIN transactions_csv...	0 row(s) returned	0.00007 sec / 0.00000...
11:04:18	SELECT company_name, phone, country, DATE(timestamp) AS fechas, amount FROM companies_csv c JOIN transactions_csv t ON...	0 row(s) returned	0.00065 sec / 0.000...
11:05:54	SELECT company_name, phone, country, DATE(timestamp) AS fechas, amount FROM companies_csv c JOIN transactions_csv t ON...	0 row(s) returned	0.0010 sec / 0.00000...
11:07:06	SELECT company_name, phone, country, DATE(timestamp), amount FROM companies_csv c JOIN transactions_csv t ON c.id = t.busi...	6 row(s) returned	0.058 sec / 0.000011...
11:07:30	SELECT company_name, phone, country, DATE(timestamp), amount FROM companies_csv c JOIN transactions_csv t ON c.id = t.busi...	6 row(s) returned	0.046 sec / 0.000010...
11:07:50	SELECT company_name, phone, country, DATE(timestamp), amount FROM companies_csv c JOIN transactions_csv t ON c.id = t.busi...	Error Code: 1064. You have an error in your SQL synta...	0.00050 sec
11:08:04	SELECT company_name, phone, country, DATE(timestamp), amount FROM companies_csv c JOIN transactions_csv t ON c.id = t.busi...	6 row(s) returned	0.046 sec / 0.000010...
11:08:29	SELECT company_name, phone, country, DATE(timestamp) AS fechas, amount FROM companies_csv c JOIN transactions_csv t ON...	6 row(s) returned	0.045 sec / 0.000010...

Versión inicial sin usar el declined = 0. Los resultados no cambian. La columna de fechas no es el alias final que se ha usado (todo ya unificado en inglés en la versión final).

Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

```
185 -- S2.N2.03 Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi,
186 -- per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses,
187 -- però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.
188
189 • SELECT c.id, company_name, COUNT(t.id) AS num_transactions,
190 CASE
191 WHEN COUNT(t.id) >= 400 THEN '400 or more'
192 ELSE 'Less than 400'
193 END AS 'Transactions'
194 FROM companies_csv c
195 LEFT JOIN transactions_csv t
196 ON c.id = t.business_id
197 GROUP BY 1
198 ORDER BY 3 ASC;
199
```

id	company_name	num_transactions	Transactions
b-2362	Lorem Eu Incorporated	380	Less than 400
b-2286	Fringilla LLC	397	Less than 400
b-2470	Nec Luctus LLC	399	Less than 400
b-2578	Dui Quis Institute	402	400 or more
b-2446	Ribus Associates	407	400 or more
b-2226	Magna A Neque Industries	410	400 or more
b-2526	Amet Luctus Vulputate Foundation	411	400 or more
b-2590	Euismod Mauris Institute	418	400 or more
b-2534	Mattis Foundation	418	400 or more
b-2298	Elit Etiam Laoreet Associates	418	400 or more

Time	Action	Response	Duration / Fetch Time
11:10:12	SELECT company_name, phone, country, DATE(timestamp) AS fechas, amount FROM company...	0 row(s) returned	0.00007 sec / 0.00000...
11:11:15	SELECT DATE(timestamp) AS fecha, SUM(amount) AS top_ingresos FROM transactions_csv GROUP BY 1 ORDER BY 2 DESC LIMIT 5	5 row(s) returned	0.070 sec / 0.000005...
11:12:00	SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS top_ingresos FROM transactions_csv GROUP BY 1 ORDER BY 2 D...	5 row(s) returned	0.066 sec / 0.00000...
11:22:24	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c JOIN transactions_csv t ON c.id = t.busine...	100 row(s) returned	0.085 sec / 0.000030...
11:24:09	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c JOIN transactions_csv t ON c.id = t.busine...	100 row(s) returned	0.045 sec / 0.000023...
12:25:47	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c LEFT JOIN transactions_csv t ON c.id = t...	100 row(s) returned	0.046 sec / 0.000032...
12:30:30	SELECT c.id, company_name, COUNT(t.id) AS num_transactions, CASE WHEN COUNT(t.id) >= 400 THEN '400 or more transaction...	100 row(s) returned	0.045 sec / 0.000020...
12:44:45	SELECT c.id, company_name, COUNT(t.id) AS num_transactions, CASE WHEN COUNT(t.id) >= 400 THEN '400 or more' ELSE 'Le...	100 row(s) returned	0.066 sec / 0.000062...

Para mostrar la consulta de forma específica vamos a introducir el CASE y así obtener una columna extra personalizada según el número de transacciones realizadas por cada empresa. Para tener un extra de claridad lo ordenamos de forma ascendiente en función de las transacciones totales y así saber rápido qué empresas realizan menos.

Exercici 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

```

1 • SELECT *
2 FROM ecommerce.transactions_csv
3 WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';

```

id	card_id	business_id	timestamp	amount	declined	product_id	user_id	lat	longitude	discount_amo...	tax_amount	shipping_amo...
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD	CcS-5019	b-2370	2016-12-21 20:07:18	172.56	0	66, 69, 87	438	41.59720554463741	12.22175994259365	19.92	29.86	6.99

transactions_csv 2

Action Output

Time	Action	Response	Duration / Fetch Time
11:11:10	SELECT DATE(timestamp) AS fecha, SUM(amount) AS top_ingresos FROM transactions_csv GROUP BY 1 ORDER BY 2 DESC LIMIT 5	5 row(s) returned	0.0000 sec / 0.000000...
11:12:00	SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS top_ingresos FROM transactions_csv GROUP BY 1 ORDER BY 2 D...	5 row(s) returned	0.066 sec / 0.000000...
11:22:24	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c JOIN transactions_csv t ON c.id = t.busine...	100 row(s) returned	0.085 sec / 0.000030...
11:24:09	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c JOIN transactions_csv t ON c.id = t.busine...	100 row(s) returned	0.045 sec / 0.000023...
12:25:47	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c LEFT JOIN transactions_csv t ON c.id = t...	100 row(s) returned	0.046 sec / 0.000032...
12:30:30	SELECT c.id, company_name, COUNT(t.id) AS num_transactions, CASE WHEN COUNT(t.id) >= 400 THEN '400 or more transaction...	100 row(s) returned	0.045 sec / 0.000020...
12:44:45	SELECT c.id, company_name, COUNT(t.id) AS num_transactions, CASE WHEN COUNT(t.id) >= 400 THEN '400 or more' ELSE 'Le...	100 row(s) returned	0.066 sec / 0.000062...
12:49:48	SELECT * FROM ecommerce.transactions_csv WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) returned	0.0020 sec / 0.00001...

Comprobamos que existe el id en la tabla "transactions_csv".

```

200
201 -- S2.N2.04 Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.
202
203 • DELETE
204 FROM transactions_csv
205 WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
206
207

```

Action Output

Time	Action	Response	Duration / Fetch Time
11:12:00	SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS top_ingresos FROM transactions_csv GROUP BY 1 ORDER BY 2 D...	5 row(s) returned	0.0000 sec / 0.000000...
11:22:24	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c JOIN transactions_csv t ON c.id = t.busine...	100 row(s) returned	0.085 sec / 0.000030...
11:24:09	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c JOIN transactions_csv t ON c.id = t.busine...	100 row(s) returned	0.045 sec / 0.000023...
12:25:47	SELECT c.id, company_name, COUNT(t.id) AS num_transactions FROM companies_csv c LEFT JOIN transactions_csv t ON c.id = t...	100 row(s) returned	0.046 sec / 0.000032...
12:30:30	SELECT c.id, company_name, COUNT(t.id) AS num_transactions, CASE WHEN COUNT(t.id) >= 400 THEN '400 or more transaction...	100 row(s) returned	0.045 sec / 0.000020...
12:44:45	SELECT c.id, company_name, COUNT(t.id) AS num_transactions, CASE WHEN COUNT(t.id) >= 400 THEN '400 or more' ELSE 'Le...	100 row(s) returned	0.066 sec / 0.000062...
12:49:48	SELECT * FROM ecommerce.transactions_csv WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) returned	0.0020 sec / 0.00001...
13:14:58	DELETE FROM transactions_csv WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) affected	0.020 sec

Ejecutamos el DELETE con el id correspondiente.

Exercici 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada

VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

```

207
208 -- S2.N2.05 La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives.
209 -- S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions.
210 -- Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació:
211 -- Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia.
212 -- Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.
213
214 • CREATE OR REPLACE VIEW vista_marketing AS
215 SELECT company_name, phone, country, AVG(amount) AS compra_media
216 FROM companies_csv c
217 LEFT JOIN transactions_csv t
218 ON c.id = t.business_id
219 GROUP BY company_name
220 ORDER BY AVG(amount) DESC;
221
222

```

Time	Action	Response	Duration / Fetch Time
13:19:48	SELECT * FROM ecommerce.transactions_csv WHERE card_id = 'Ccs-5019'	20 row(s) returned	0.0003 sec / 0.0003...
13:19:47	SELECT * FROM ecommerce.transactions_csv WHERE card_id = 'Ccs-5019' AND amount = 172.56	0 row(s) returned	0.0013 sec / 0.0001...
13:19:56	SELECT * FROM ecommerce.transactions_csv WHERE card_id = 'Ccs-5019' AND amount = 172.56	0 row(s) returned	0.0016 sec / 0.0002...
13:36:38	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) FROM companies_csv c LEFT JOIN transa...	Error Code: 1054. Unknown column 't.business_id' in...	0.0080 sec
13:36:55	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) FROM companies_csv c LEFT JOIN transa...	0 row(s) affected	0.016 sec
13:37:56	SELECT * FROM ecommerce.transactions_csv	99999 row(s) returned	0.0014 sec / 0.100 sec
13:41:04	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) AS compra_media FROM companies_csv...	Error Code: 1050. Table 'vista_marketing' already exists	0.0040 sec
13:42:28	CREATE OR REPLACE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) AS compra_media FROM co...	0 row(s) affected	0.0064 sec

Hemos creado una complex view llamada vista_marketing con los campos solicitados. Después de haberlo ejecutado me doy cuenta de que no lo tenía que poner aquí si no en el apartado de "Views" (eso pasa por no ver TODA la documentación completa antes del ejercicio).

Name: marketing_vista

```

1 • CREATE
2     ALGORITHM = UNDEFINED
3     DEFINER = 'root'@'localhost'
4     SQL SECURITY DEFINER
5     VIEW `marketing_vista` AS
6     SELECT
7         `c`.`company_name` AS `company_name`,
8         `c`.`phone` AS `phone`,
9         `c`.`country` AS `country`,
10        ROUND(AVG(`t`.`amount`), 2) AS `compra_media`
11    FROM
12        (`companies_csv` `c`
13     LEFT JOIN `transactions_csv` `t` ON ((`c`.`id` = `t`.`business_id`)))
14    GROUP BY `c`.`company_name`, `c`.`phone`, `c`.`country`
15    ORDER BY AVG(`t`.`amount`) DESC

```

Time	Action	Response	Duration / Fetch Time
13:36:38	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) FROM companies_csv c LEFT JOIN transa...	Error Code: 1054. Unknown column 't.business_id' in...	0.0080 sec
13:36:55	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) FROM companies_csv c LEFT JOIN transa...	0 row(s) affected	0.016 sec
13:37:56	SELECT * FROM ecommerce.transactions_csv	99999 row(s) returned	0.0014 sec / 0.100 sec
13:41:04	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) AS compra_media FROM companies_csv...	Error Code: 1050. Table 'vista_marketing' already e...	0.0040 sec
13:42:28	CREATE OR REPLACE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) AS compra_media FROM co...	0 row(s) affected	0.0064 sec
13:50:16	Apply changes to vista_marketing	Selected name conflicts with existing view 'vista_m...	
13:50:45	Apply changes to vista_marketing	Selected name conflicts with existing view 'vista_m...	
13:51:14	Apply changes to marketing_vista	Changes applied	
13:51:46	SELECT * FROM ecommerce.marketing_vista	Error Code: 1055. Expression #2 of SELECT list is n...	0.00069 sec
13:51:55	SELECT * FROM ecommerce.marketing_vista	Error Code: 1055. Expression #2 of SELECT list is n...	0.00038 sec
13:54:38	SELECT * FROM ecommerce.marketing_vista	Error Code: 1055. Expression #2 of SELECT list is n...	0.0017 sec
13:56:11	Apply changes to marketing_vista	Changes applied	
13:56:22	SELECT * FROM ecommerce.marketing_vista	100 row(s) returned	0.225 sec / 0.000028...
13:56:51	Apply changes to marketing_vista	Changes applied	
13:56:59	SELECT * FROM ecommerce.marketing_vista	100 row(s) returned	0.197 sec / 0.000049...

Una vez visto el tutorial corto y sencillo... Creamos la vista "marketing_vista" con los campos especificados y con un orden de mayor a menor media de compra de cada empresa. La vista para marketing se vería de este modo:

1	•	SELECT * FROM ecommerce.marketing_vista;
---	---	--

company_name	phone	country	compra_media
Ac Fermentum Incorporated	+64 601 244 6320	Germany	330.77
Ut Semper Foundation	+10 283 500 6448	Sweden	307.37
Orci Adipiscing Limited	+60 897 390 5820	United Kingdom	306.27
Dolor Vitae Limited	+6 421 567 9195	France	305.95
Nunc In Foundation	+13 267 741 2819	Italy	305.4
Maecenas Malesuada Fringilla Inc.	+51 673 540 5289	Netherlands	305.15
Aliquet Sem Limited	+44 744 425 9915	Netherlands	304.21
Nulla Integer Vulputate Corp.	+75 236 687 9162	Sweden	303.42
Non Justo Corp.	+67 963 294 1566	Sweden	303.3
Turpis Company	+17 624 311 7168	Netherlands	302.77
Aliquam Erat Volutpat LLP	+39 372 330 7229	Italy	301.57
Amet Lorem LLP	+40 468 847 7738	Spain	301.82
Placerat LLP	+17 324 602 5033	Netherlands	301.61
Tempor Diam Institute	+89 848 141 3136	Netherlands	301.6
Integer Mollis Corp.	+38 803 863 7881	Italy	301.29
Nunc Interdum Incorporated	+54 217 870 5363	Germany	300.69
Etiam Rhoncus Inc. Fermentum Inc.	+70 840 478 2460	France	300.36

Time	Action	Response	Duration / Fetch Time
44 13:36:38	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) FROM companies_csv c LEFT JOIN transa...	Error Code: 1054. Unknown column 't.business_id' in 'field list'	0.0080 sec
45 13:36:55	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) FROM companies_csv c LEFT JOIN transa...	0 row(s) affected	0.016 sec
46 13:37:56	SELECT * FROM ecommerce.transactions_csv	99999 row(s) returned	0.0014 sec / 0.100 sec
47 13:41:04	CREATE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) AS compra_media FROM companies_csv...	Error Code: 1050. Table 'vista_marketing' already exists	0.0040 sec
48 13:42:28	CREATE OR REPLACE VIEW vista_marketing AS SELECT company_name, phone, country, AVG(amount) AS compra_media FROM co...	0 row(s) affected	0.0064 sec
49 13:50:16	Apply changes to vista_marketing	Selected name conflicts with existing view 'vista_m...	
50 13:50:45	Apply changes to vista_marketing	Selected name conflicts with existing view 'vista_m...	
51 13:51:14	Apply changes to marketing_vista	Changes applied	
52 13:51:46	SELECT * FROM ecommerce.marketing_vista	Error Code: 1055. Expression #2 of SELECT list is n...	0.00069 sec
53 13:51:55	SELECT * FROM ecommerce.marketing_vista	Error Code: 1055. Expression #2 of SELECT list is n...	0.00038 sec
54 13:54:38	SELECT * FROM ecommerce.marketing_vista	Error Code: 1055. Expression #2 of SELECT list is n...	0.0017 sec
55 13:56:11	Apply changes to marketing_vista	Changes applied	
56 13:56:22	SELECT * FROM ecommerce.marketing_vista	100 row(s) returned	0.225 sec / 0.000028...
57 13:56:51	Apply changes to marketing_vista	Changes applied	
58 13:56:59	SELECT * FROM ecommerce.marketing_vista	100 row(s) returned	0.197 sec / 0.000049...

▼ Nivell 3 ★★★★★

Exercici 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:

222	--	SPRINT 2 - NIVELL 3
223	--	Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat
224	--	declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:
225	--	Quantes targetes estan actives?
226	--	
227	--	
228	•	SELECT card_id, declined, DATE(timestamp) AS fecha
229		FROM transactions_csv t
230		WHERE card_id = 'CcS-7639'
231		ORDER BY DATE(timestamp) DESC
232		LIMIT 3;
233		
234		

card_id	declined	fecha
CcS-7639	0	2024-06-12
CcS-7639	0	2024-03-14
CcS-7639	0	2023-08-11

Time	Action	Response	Duration / Fetch Time
1 10:36:45	SELECT * FROM transactions.credit_card	5001 row(s) returned	0.0075 sec / 0.0070 s...
2 11:05:02	SELECT * FROM transactions.transaction	2573 row(s) returned	0.0030 sec / 0.0040...
3 11:05:42	SELECT card_id, declined FROM transactions_csv t WHERE card_id = 'CcS-7639'	15 row(s) returned	0.0052 sec / 0.00001...
4 11:07:47	SELECT card_id, declined, DATE(timestamp) AS fecha FROM transactions_csv t WHERE card_id = 'CcS-7639' ORDER BY DATE(time...	15 row(s) returned	0.0017 sec / 0.00000...
5 11:08:04	SELECT card_id, declined, DATE(timestamp) AS fecha FROM transactions_csv t WHERE card_id = 'CcS-7639' ORDER BY DATE(time...	3 row(s) returned	0.00076 sec / 0.0000...

Primero vamos a realizar una consulta simple con un único card_id para entender bien el core del enunciado.

233

234

235

236

237

100%

1:237

Result Grid

Filter Rows: Search

Export: Fetch rows:

card_id

declined

fecha

ranking1

CcS-4857

0

2024-10-25

1

CcS-4857

0

2024-10-07

2

CcS-4857

0

2024-07-27

3

CcS-4857

0

2024-02-08

4

CcS-4857

0

2024-01-15

5

CcS-4857

0

2022-09-21

6

CcS-4857

0

2022-06-20

7

CcS-4857

0

2021-10-02

8

CcS-4857

0

2021-09-30

9

CcS-4857

0

2021-06-30

10

CcS-4857

0

2021-02-14

11

CcS-4857

0

2021-01-18

12

CcS-4857

0

2021-01-11

13

CcS-4857

0

2020-10-01

14

CcS-4857

0

2020-03-13

15

CcS-4857

0

2020-01-25

16

CcS-4857

0

2019-10-22

17

Result 4

Read Only

Action Output

Time

Action

Response

Duration / Fetch Time

1

10:36:45

SELECT * FROM transactions.credit_card

5001 row(s) returned

0.0075 sec / 0.0070 s...

2

11:05:02

SELECT * FROM transactions.transaction

2573 row(s) returned

0.0030 sec / 0.0040...

3

11:05:42

SELECT card_id, declined FROM transactions_csv t WHERE card_id = 'CcS-7639'

15 row(s) returned

0.0052 sec / 0.00001...

4

11:07:47

SELECT card_id, declined, DATE(timestamp) AS fecha FROM transactions_csv t WHERE card_id = 'CcS-7639' ORDER BY DATE(timestamp) DESC

15 row(s) returned

0.0017 sec / 0.00000...

5

11:08:04

SELECT card_id, declined, DATE(timestamp) AS fecha FROM transactions_csv t WHERE card_id = 'CcS-7639' ORDER BY DATE(timestamp) DESC

3 row(s) returned

0.00076 sec / 0.00000...

6

11:33:55

SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1

99999 row(s) returned

0.100 sec / 0.022 sec

Probamos también la función de ventana en términos generales ya que es algo nuevo para mí. Comprobamos que funciona con la tabla completa de "transactions_csv" y que nos devuelve 99999 rows ya que en el nivel anterior hemos eliminado uno de los registros.

238

• SELECT card_id, declined, fecha, ranking1

239

FROM (SELECT card_id, declined, timestamp AS fecha,

240

ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp DESC) AS ranking1

241

FROM transactions_csv t) AS window_cards

242

WHERE window_cards.ranking1 IN (1, 2, 3);

243

100%

42:242

Result Grid

Filter Rows:

Export:

Fetch rows:

card_id	declined	fecha	ranking1
CcS-4875	0	2024-11-28 17:44:02	3
CcS-4876	0	2024-12-28 07:00:13	1
CcS-4876	0	2024-05-06 16:03:32	2
CcS-4876	0	2023-10-18 16:28:14	3
CcS-4877	0	2024-02-13 22:09:02	1
CcS-4877	0	2023-09-27 04:12:35	2
CcS-4877	0	2023-07-01 03:52:09	3
CcS-4878	0	2024-10-22 23:13:00	1
CcS-4878	0	2024-05-10 21:30:47	2
CcS-4878	0	2023-12-22 02:01:35	3
CcS-4879	0	2024-04-01 08:51:55	1
CcS-4879	0	2023-11-16 03:59:26	2
CcS-4879	0	2023-10-10 30:54:48	3
Result 8			

Read Only

Result Grid

Form Editor

Field Types

▼

Action Output

	Time	Action	Response
✓ 2	11:05:02	SELECT * FROM transactions.transaction	2573 row(s) returned
✓ 3	11:05:42	SELECT card_id, declined FROM transactions_csv t WHERE card_id = 'CcS-7639'	15 row(s) returned
✓ 4	11:07:47	SELECT card_id, declined, DATE(timestamp) AS fecha FROM transactions_csv t WHERE card_id = 'CcS-7639' ORDER BY DATE(timestamp) DESC	15 row(s) returned
✓ 5	11:08:04	SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	3 row(s) returned
✓ 6	11:33:55	SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	99999 row(s) returned
✓ 7	11:43:12	SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	0 row(s) returned
✗ 8	11:44:28	SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	Error Code: 1054. Unknown column 'ranking1' in 'where clause'
✗ 9	11:44:42	SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	0 row(s) returned
✗ 10	11:58:19	SELECT card_id, declined, DATE(timestamp) AS fecha FROM (SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	Error Code: 1064. You have an error in your SQL syntax; check th
✗ 11	11:58:32	SELECT card_id, declined, DATE(timestamp) AS fecha FROM (SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	Error Code: 1054. Unknown column 'timestamp' in 'field list'
✗ 12	12:00:11	SELECT card_id, declined, DATE(timestamp) AS fecha FROM (SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	Error Code: 1054. Unknown column 'timestamp' in 'field list'
✗ 13	12:01:53	SELECT card_id, declined, fecha FROM (SELECT card_id, declined, DATE(timestamp) AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	Error Code: 1064. You have an error in your SQL syntax; check th
✗ 14	12:02:52	SELECT card_id, declined, fecha FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	Error Code: 1054. Unknown column 'window_cards.ranking1' in 'w
✓ 15	12:03:02	SELECT card_id, declined, fecha FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	15000 row(s) returned
✗ 16	12:03:30	SELECT window_cards FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	Error Code: 1054. Unknown column 'window_cards' in 'field list'
✓ 17	12:06:09	SELECT card_id, declined, fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY DATE(timestamp) DESC) AS ranking1	15000 row(s) returned

Después de darnos algunos golpes para entender bien cómo filtrar solo las 3 últimas transacciones de cada tarjeta, conseguimos la consulta con la función ventana y el filtro aplicado. A continuación vamos a intentar conseguir las tarjetas activas e inactivas.

```

237
238 • SELECT card_id, SUM(declined) AS card_declined
239 FROM (SELECT card_id, declined, timestamp AS fecha,
240         ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp DESC) AS ranking1
241      FROM transactions_csv t) AS window_cards
242 WHERE window_cards.ranking1 IN (1, 2, 3)
243 GROUP BY card_id;
244

```

card_id	SUM(declined)
CcS-5415	0
CcS-5416	0
CcS-5417	0
CcS-5418	0
CcS-5419	0
CcS-5420	0
CcS-5421	0
CcS-5422	0

Result 10

Time	Action	Response
16 12:03:30	SELECT window_cards FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp...	Error Code: 1054. Unknown column 'window_cards' in 'field list'
17 12:06:09	SELECT card_id, declined, fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id OR...	15000 row(s) returned
18 12:22:07	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	Error Code: 1140. In aggregated query without GROUP BY, expr...
19 12:22:33	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	15000 row(s) returned
20 12:23:01	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	Error Code: 1055. Expression #2 of SELECT list is not in GROUP
21 12:24:51	SELECT card_id, SUM(declined) FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY ti...	5000 row(s) returned

Dejamos lista la consulta para usar el CASE y así poder indicar bien si esta activa o no mediante el sumatorio de veces que ha sido "declined".

```

237
238 • SELECT card_id, SUM(declined) AS card_declined,
239 CASE
240 WHEN SUM(declined) = 3 THEN 'Inactive'
241 ELSE 'Active'
242 END AS 'card_status'
243 FROM (SELECT card_id, declined, timestamp AS fecha,
244         ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp DESC) AS ranking1
245      FROM transactions_csv t) AS window_cards
246 WHERE window_cards.ranking1 IN (1, 2, 3)
247 GROUP BY card_id;
248

```

card_id	card_declined	card_status
CcS-4857	0	Active
CcS-4858	0	Active
CcS-4859	0	Active
CcS-4860	0	Active
CcS-4861	0	Active
CcS-4862	0	Active
CcS-4863	0	Active
CcS-4864	0	Active

Result 11

Time	Action	Response
17 12:06:09	SELECT card_id, declined, fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id OR...	15000 row(s) returned
18 12:22:33	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	Error Code: 1140. In aggregated query without GROUP BY, expr...
19 12:23:01	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	15000 row(s) returned
21 12:24:51	SELECT card_id, SUM(declined) FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY ti...	5000 row(s) returned
22 12:30:49	SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'Inactive' ELSE 'Active' END AS 'card_status' FROM (SELE...	5000 row(s) returned

Ya tenemos nuestra consulta con la información del estado de las tarjetas de crédito activas o inactivas.

```

223 -- SPRINT 2 - NIVELL 3
224 -- Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat
225 -- declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon:
226 -- Quantes targetes estan actives?
227
228 • DROP TABLE IF EXISTS credit_card_status;
229
230 • CREATE TABLE credit_card_status AS
231 SELECT card_id, SUM(declined) AS card_declined,
232 CASE
233 WHEN SUM(declined) = 3 THEN 'Inactive'
234 ELSE 'Active'
235 END AS 'card_status'
236 FROM (SELECT card_id, declined, timestamp AS fecha,
237 ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY timestamp DESC) AS ranking1
238 FROM transactions_csv t) AS window_cards
239 WHERE window_cards.ranking1 IN (1, 2, 3)
240 GROUP BY card_id;
241

```

100% 1:230

Action Output

	Time	Action	Response
19	12:22:33	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	15000 row(s) returned
20	12:23:01	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	Error Code: 1055: Expression #2 of SELECT list is not in GROUP...
21	12:24:51	SELECT card_id, SUM(declined) FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY t...	5000 row(s) returned
22	12:30:49	SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'Inactive' ELSE 'Active' END AS 'card_status' FROM (SELE...	5000 row(s) returned
23	12:48:49	DROP TABLE IF EXISTS credit_card_status	0 row(s) affected, 1 warning(s): 1051 Unknown table 'ecommerce
24	12:49:02	CREATE TABLE credit_card_status AS SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'Inactive' ELSE 'Acti...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0

Creemos nuestra tabla "credit_card_status" con las condiciones y características previamente comprobadas. A continuación vemos nuestra nueva tabla:

1 • SELECT * FROM ecommerce.credit_card_status;

100% 1:1

Result Grid Filter Rows: Search Export: Fetch rows:

card_id	card_declined	card_status
CCS-4857	0	Active
CCS-4858	0	Active
CCS-4859	0	Active
CCS-4860	0	Active
CCS-4861	0	Active
CCS-4862	0	Active
CCS-4863	0	Active
CCS-4864	0	Active
CCS-4865	0	Active
CCS-4866	0	Active
CCS-4867	0	Active
CCS-4868	0	Active
CCS-4869	0	Active
CCS-4870	0	Active
CCS-4871	0	Active
CCS-4872	0	Active
CCS-4873	0	Active
CCS-4874	0	Active
CCS-4875	0	Active
CCS-4876	0	Active
CCS-4877	0	Active
CCS-4878	0	Active
CCS-4879	0	Active
CCS-4880	0	Active
CCS-4881	0	Active
CCS-4882	0	Active

credit_card_status 1

Action Output

	Time	Action	Response
20	12:23:01	SELECT card_id, declined, SUM(declined), fecha, ranking1 FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION...	Error Code: 1055: Expression #2 of SELECT list is not in GROUP...
21	12:24:51	SELECT card_id, SUM(declined) FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY t...	5000 row(s) returned
22	12:30:49	SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'Inactive' ELSE 'Active' END AS 'card_status' FROM (SELE...	5000 row(s) returned
23	12:48:49	DROP TABLE IF EXISTS credit_card_status	0 row(s) affected, 1 warning(s): 1051 Unknown table 'ecommerce
24	12:49:02	CREATE TABLE credit_card_status AS SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'Inactive' ELSE 'Acti...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0
25	12:50:23	SELECT * FROM ecommerce.credit_card_status	5000 row(s) returned

👉 Quantes targetes estan actives?

```

240
241 -- Quantes targetes estan actives?
242 • SELECT card_status, COUNT(card_status) AS total
243 FROM credit_card_status
244 GROUP BY card_status;
245

```

100% 22:244

Result Grid Filter Rows: Search Export:

card_status	total
Active	5000

Result 12 Read Only

Action Output

	Time	Action	Response
✓	21	12:24:51 SELECT card_id, SUM(declined) FROM (SELECT card_id, declined, timestamp AS fecha, ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY ti...	5000 row(s) returned
✓	22	12:30:49 SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'inactive' ELSE 'Active' END AS 'card_status' FROM (SELE...	5000 row(s) returned
⚠	23	12:48:49 DROP TABLE IF EXISTS credit_card_status	0 row(s) affected, 1 warning(s): 1051 Unknown table 'ecommerce
✓	24	12:49:02 CREATE TABLE credit_card_status AS SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'inactive' ELSE 'Acti...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0
✓	25	12:50:23 SELECT * FROM ecommerce.credit_card_status	5000 row(s) returned
✓	26	12:54:28 SELECT card_status, COUNT(card_status) AS total FROM credit_card_status GROUP BY card_status	1 row(s) returned

El número de tarjetas activas es 5000, es decir que todas las tarjetas están activas según el criterio de las últimas 3 transacciones. Para comprobarlo, vemos cuál es el máximo valor de veces que se ha declinado la tarjeta en las 3 últimas transacciones y vemos que es 2. No se llega a 3 en ninguno de estos caso. Lo observamos en la siguiente imagen:

```

1 • SELECT *
2 FROM ecommerce.credit_card_status
3 ORDER BY card_declined DESC;

```

100% 28:3

Result Grid Filter Rows: Search Export: Fetch rows:

card_id	card_declined	card_status
CcS-9406	2	Active
CcS-4891	1	Active
CcS-4939	1	Active
CcS-4957	1	Active
CcS-4969	1	Active
CcS-5006	1	Active
CcS-5021	1	Active
CcS-5090	1	Active
CcS-5110	1	Active
CcS-5266	1	Active
CcS-5346	1	Active
CcS-5377	1	Active

credit_card_status 2 Read Only

Action Output

	Time	Action	Response
✓	22	12:30:49 SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'inactive' ELSE 'Active' END AS 'card_status' FROM (SELE...	5000 row(s) returned
⚠	23	12:48:49 DROP TABLE IF EXISTS credit_card_status	0 row(s) affected, 1 warning(s): 1051 Unknown table 'ecommerce
✓	24	12:49:02 CREATE TABLE credit_card_status AS SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'inactive' ELSE 'Acti...	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0
✓	25	12:50:23 SELECT * FROM ecommerce.credit_card_status	5000 row(s) returned
✓	26	12:54:28 SELECT card_status, COUNT(card_status) AS total FROM credit_card_status GROUP BY card_status	1 row(s) returned
✓	27	12:57:00 SELECT * FROM ecommerce.credit_card_status ORDER BY card_declined DESC	5000 row(s) returned

Exercici 2

Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podem fer-ho), tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

N1-Ex.8_products

Visualización Zoom Añadir categoría Tabla de pivotación Insertar Tabla Gráfica Texto Figura Multime

Hoja 1

N1-Ex.8_products

id	product_name	price	colour	weight	warehouse_id	category	brand	cost	launch_date
1	Direwolf Stannis	\$161.11	#7c7c7c	1.0	WH-4	Sports	Runway Pro	\$63.84	2014-11-01
2	Tarly Stark	\$9.24	#919191	2.0	WH-3	Gaming	Questbyte	\$4.74	2013-02-07
3	duel tourney Lannister	\$171.13	#d8d8d8	1.5	WH-2	Food	Daily Pantry	\$107.07	2013-08-18
4	warden south duel	\$71.89	#111111	3.0	WH-1	Books	NorthPress	\$39.17	2013-12-31
5	skywalker ewok	\$171.22	#dbdbdb	3.2	WH-0	Home	NordHaus	\$83.69	2014-07-17
6	dooku solo	\$136.60	#c4c4c4	0.8	WH-1	Home	NordHaus	\$66.78	2014-03-14
7	north of Caste								-10-10
8	Winterfell								-01-27
9	Winterfell								-08-14
10	Karstark Dorn								-10-04
11	Karstark Dorn								-09-01
12	duel Direwolf								-10-11
13	palpatine che								-04-26
14	Direwolf								-03-29
15	Stannis warde								-10-21

N1-Ex.8_products.csv

id,product_name,price,colour,weight,warehouse_id,category,brand,cost,launch_date

1,Direwolf Stannis,\$161.11,#7c7c7c,1.0,WH-4,Sports,Runway Pro,\$63.84,2014-11-01

2,Tarly Stark,\$9.24,#919191,2.0,WH-3,Gaming,Questbyte,\$4.74,2013-02-07

3,duel tourney Lannister,\$171.13,#d8d8d8,1.5,WH-2,Food,Daily Pantry,\$107.07,2013-08-18

4,warden south duel,\$71.89,#111111,3.0,WH-1,Books,NorthPress,\$39.17,2013-12-31

5,skywalker ewok,\$171.22,#dbdbdb,3.2,WH-0,Home,NordHaus,\$83.69,2014-07-17

6,dooku solo,\$136.60,#c4c4c4,0.8,WH-1,Home,NordHaus,\$66.78,2014-03-14

7,north of Casterly,\$63.33,#b7b7b7,0.6,WH-2,Electronics,Nexa,\$37.41,2013-10-10

8,Winterfell,\$32.37,#383838,1.4,WH-3,Travel,Voyago,\$15.20,2013-01-27

9,Winterfell,\$76.40,#b5b5b5,1.2,WH-4,Books,Campus Read,\$40.42,2013-08-14

10,Karstark Dorne,\$119.52,#f4f4f4,2.4,WH-5,Food,FreshBox,\$89.07,2013-10-04

11,Karstark Dorne,\$49.70,#141414,2.7,WH-6,Electronics,Nexa,\$34.45,2014-09-01

12,duel Direwolf,\$181.60,#a8a8a8,2.1,WH-7,Gaming,Questbyte,\$99.46,2013-10-11

13,palpatine chewbacca,\$139.59,#2b2b2b,1.0,WH-8,Travel,CityHop,\$82.76,2014-04-26

Antes de hacer nada vamos a observar el csv que tenemos que añadir para ver qué campos tiene y saber como se separan (con coma en este caso).

```

247 -- S2.W3.02 Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada
248 -- (ja que fins ara no podem fer-ho), tenint en compte que des de transaction tens product_ids. Genera la següent consulta:
249
250 * CREATE TABLE IF NOT EXISTS products_csv (
251     id VARCHAR(255) PRIMARY KEY,
252     product_name VARCHAR(255) NULL,
253     price VARCHAR(255) NULL,
254     colour VARCHAR(255) NULL,
255     weight VARCHAR (255) NULL,
256     warehouse_id VARCHAR(255) NULL,
257     category VARCHAR(255) NULL,
258     brand VARCHAR(255) NULL,
259     cost VARCHAR (15) NULL,
260     launch_date VARCHAR (255) NULL
261 );
262
100% 3:261
Action Output
Time Action Response
23 12:48:49 DROP TABLE IF EXISTS credit_card_status 0 row(s) affected, 1 warning(s): 1051 Unknown table 'ecommerc
24 12:49:02 CREATE TABLE credit_card_status AS SELECT card_id, SUM(declined) AS card_declined, CASE WHEN SUM(declined) = 3 THEN 'Inactive' ELSE 'Acti... 6000 row(s) affected Records: 5000 Duplicates: 0 Warnings: C
25 12:50:23 SELECT * FROM ecommerce.credit_card_status 5000 row(s) returned
26 12:54:28 SELECT card_status, COUNT(card_status) AS total FROM credit_card_status GROUP BY card_status 1 row(s) returned
27 12:57:00 SELECT * FROM ecommerce.credit_card_status ORDER BY card_declined DESC 5000 row(s) returned
28 13:32:55 CREATE TABLE IF NOT EXISTS products_csv ( id VARCHAR(255) PRIMARY KEY, product_name VARCHAR(255) NULL, price VARCHAR(255) NU... 0 row(s) affected

```

Creemos la tabla "products_csv" vacía y de forma générica.

264	
265	LOAD DATA
266	LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD/N1-Ex.8__products.csv'
267	INTO TABLE products_csv
268	FIELDS TERMINATED BY ','
269	ENCLOSED BY ''
270	IGNORE 1 ROWS;
271	

100%	16:269
------	--------

Time	Action	Response	Duration / Fetch Time
12:57:00	SELECT * FROM ecommerce.credit_card_status ORDER BY card_declined DESC	5000 row(s) returned	0.0055 sec / 0.0011 s...
13:32:55	CREATE TABLE IF NOT EXISTS products_csv (id VARCHAR(255) PRIMARY KEY, product_name VARCHAR...	0 row(s) affected	0.032 sec
13:34:18	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD...	Error Code: 3948. Loading local data is disabled; this must be enabled on both t...	0.0021 sec
13:34:38	SET GLOBAL local_infile = 1	0 row(s) affected	0.0014 sec
13:34:42	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD...	100000 row(s) affected, 1024 warning(s): 1261 Row 1 doesn't contain data for a...	1.180 sec
13:36:06	SELECT * FROM ecommerce.credit_card_status	5000 row(s) returned	0.0015 sec / 0.0043...
13:36:22	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.0069 sec
13:36:30	SELECT * FROM ecommerce.products_csv	100100 row(s) returned	0.00076 sec / 0.066...
13:36:39	DROP TABLE IF EXISTS products_csv	0 row(s) affected	0.013 sec
13:38:45	CREATE TABLE IF NOT EXISTS products_csv (id VARCHAR(255) PRIMARY KEY, product_name VARCHAR...	0 row(s) affected	0.0097 sec
13:38:57	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.0025 sec
13:39:04	SELECT * FROM ecommerce.products_csv	100 row(s) returned	0.00072 sec / 0.0000...

Cargamos los datos del cvs de forma local como en ejercicios anteriores. En caso de mirar el Action Output podremos observar algún pequeño error ya que sin querer copié la ruta de un csv equivocado... pequeño error que se corrige añadiendo el DROP TABLE IF EXIST al principio de la consulta para correr un tupido velo y crear la tabla de nuevo con la carga correcta de datos de productos. Con esto ya tenemos nuestra nueva tabla con los datos:

1	SELECT * FROM ecommerce.products_csv;
---	---------------------------------------

100%	1:1
------	-----

id	product_name	price	colour	weight	warehouse_id	category	brand	cost	launch_date
1	Direwolf Stannis	\$161.11	#7c7c7c	1.0	WH-4	Sports	Runway Pro	\$63.84	2014-11-01
10	Karstark Dome	\$119.52	#4d4d4d	2.4	WH-5	Food	FreshBox	\$89.07	2013-10-04
100	south duel	\$40.43	#6b6b6b	3.0	WH-95	Electronics	PixelForge	\$24.95	2013-01-05
11	Karstark Dome	\$49.70	#f1f1f1	2.7	WH-6	Electronics	Nexa	\$34.45	2014-09-01
12	duel Direwolf	\$181.60	#a8a8a8	2.1	WH-7	Gaming	Questbyte	\$99.46	2013-10-11
13	palpatine chowbaacca	\$136.59	#2b2b2b	1.0	WH-8	Travel	CityHop	\$82.76	2014-04-26
14	Direwolf	\$147.53	#c4c4c4	2.0	WH-9	Home	NordHaus	\$71.07	2014-03-29
15	Stannis warden	\$194.29	#dbdbdb	1.5	WH-10	Home	CasaNova	\$107.93	2014-10-21
16	the duel warden	\$180.91	#666666	3.0	WH-11	Home	NordHaus	\$101.11	2014-05-31
17	skywalker ewok slith	\$91.89	#7c7c7c	3.2	WH-12	Fashion	Arden	\$38.01	2014-07-01
18	Karstark warden	\$148.91	#c4c4c4	0.8	WH-13	Beauty	CamelLab	\$47.62	2013-07-11
19	dcou solo	\$60.33	#3f3f3f	0.6	WH-14	Sports	Trail&Co	\$27.81	2013-02-18
2	Tarly Stark	\$9.24	#019191	2.0	WH-3	Gaming	Questbyte	\$4.74	2013-02-07
20	warden Karstark	\$91.96	#5b5b5b	1.4	WH-15	Fashion	Blue Loom	\$40.04	2014-05-27
21	duel Direwolf	\$96.90	#e2e2e2	1.2	WH-16	Books	PageWorks	\$62.72	2013-01-17
22	chowbaacca mustafar	\$150.02	#f1f1f1	2.4	WH-17	Home	NordHaus	\$89.46	2013-06-21
23	riverlands north	\$169.96	#545454	2.7	WH-18	Fashion	Arden	\$55.00	2013-08-08
24	south duel tourney	\$48.99	#a8a8a8	2.1	WH-19	Sports	MotionLab	\$24.26	2013-05-07
25	skywalker ewok	\$157.53	#2b2b2b	1.0	WH-20	Beauty	Luma Skin	\$53.56	2014-04-22
26	Stark Karstark	\$53.01	#898989	2.0	WH-21	Food	GreenFork	\$30.49	2013-02-15
27	Stannis riverlands	\$172.93	#7a7a7a	1.5	WH-22	Home	NordHaus	\$83.83	2014-01-08
28	chowbaacca mustafar	\$127.44	#efefef	3.0	WH-23	Beauty	Luma Skin	\$45.65	2014-06-02
29	Tully maester Tarly	\$167.20	#111111	3.2	WH-24	Books	PageWorks	\$92.89	2013-04-09
3	duel tourney Lannis...	\$171.13	#4b4b4b	1.5	WH-2	Food	Daily Pantry	\$107.07	2013-08-18
30	Karstark warden	\$79.53	#606060	0.8	WH-25	Electronics	Nexa	\$56.18	2013-02-06
31	Lannister	\$85.02	#3f3f3f	0.6	WH-26	Books	Campus R...	\$57.38	2013-10-07
99	south	\$178.28	#010101	1.4	WH-27	Home	CasaNova	\$68.47	2013-12-15

products_csv 1	Apply	Revert
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Time	Action	Response	Duration / Fetch Time
13:38:39	DROP TABLE IF EXISTS products_csv	0 row(s) affected	0.013 sec
13:38:45	CREATE TABLE IF NOT EXISTS products_csv (id VARCHAR(255) PRIMARY KEY, product_name VARCHAR...	0 row(s) affected	0.0097 sec
13:38:57	LOAD DATA LOCAL INFILE '/Users/laurea/Desktop/CURSO_FADD/1_Especialització/1.3_Sprints/Sprint_2/BBDD...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0	0.0025 sec
13:39:04	SELECT * FROM ecommerce.products_csv	100 row(s) returned	0.00072 sec / 0.0000...

Tabla "products_csv" con los datos cargados.

271
272 • `SELECT FIND_IN_SET('73', '16, 26, 97, 87');`
273

100% 15:270

Result Grid Filter Rows: Search Export:

FIND_IN_SET('73', '16, 26, 97, 87...)
0

Result 1 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 2	00:01:35	SELECT * FROM ecommerce.products_csv	100 row(s) returned	0.0020 sec / 0.0000...
✓ 3	01:06:01	SELECT * FROM transactions.transaction	2573 row(s) returned	0.0024 sec / 0.0035...
✓ 4	01:06:45	SELECT * FROM ecommerce.transactions_csv	99999 row(s) returned	0.0024 sec / 0.148 sec
✓ 5	01:08:31	SELECT FIND_IN_SET('73', '16, 26, 97, 87')	1 row(s) returned	0.00073 sec / 0.0000...

Probamos el FIND_IN_SET para entender cómo funciona ya que la tabla de transacciones almacena una lista en vez de un único id de producto como con el resto de los campos.

271
272 • `SELECT FIND_IN_SET('87', '16, 26, 97, 87');`
273

100% 16:269

Result Grid Filter Rows: Search Export:

FIND_IN_SET('87', '16, 26, 97, 87...)
0

Result 2 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 3	01:06:01	SELECT * FROM transactions.transaction	2573 row(s) returned	0.0024 sec / 0.0035...
✓ 4	01:06:45	SELECT * FROM ecommerce.transactions_csv	99999 row(s) returned	0.0024 sec / 0.148 sec
✓ 5	01:08:31	SELECT FIND_IN_SET('73', '16, 26, 97, 87')	1 row(s) returned	0.00073 sec / 0.0000...
✓ 6	01:11:04	SELECT FIND_IN_SET('87', '16, 26, 97, 87')	1 row(s) returned	0.00059 sec / 0.0000...

Me encuentro con que no funciona ya que me devuelve 0 cuando debería salir 4...

271
272 • `SELECT FIND_IN_SET('87', REPLACE('16, 26, 97, 87', ' ', ''));`
273

100% 15:270

Result Grid Filter Rows: Search Export:

FIND_IN_SET('87', REPLACE('16, 26, 97, 87', ' ', ''))
4

Result 3 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 4	01:06:45	SELECT * FROM ecommerce.transactions_csv	99999 row(s) returned	0.0024 sec / 0.148 sec
✓ 5	01:08:31	SELECT FIND_IN_SET('73', '16, 26, 97, 87')	1 row(s) returned	0.00073 sec / 0.0000...
✓ 6	01:11:04	SELECT FIND_IN_SET('87', '16, 26, 97, 87')	1 row(s) returned	0.00059 sec / 0.0000...
✓ 7	01:13:28	SELECT FIND_IN_SET('87', REPLACE('16, 26, 97, 87', ' ', ''))	1 row(s) returned	0.00088 sec / 0.0000...

Añadimos el REPLACE y ahora sí nos da el 4 indicando la posición correcta en la lista.

273
274 • `DROP TABLE IF EXISTS transactions_products;`
275
276 • `CREATE TABLE transactions_products AS`
277 `SELECT t.id AS transaction_id,`
278 `p.id AS product_id`
279 `FROM transactions_csv t`
280 `JOIN products_csv p`
281 `ON FIND_IN_SET(p.id, REPLACE(t.product_ids, ' ', '')) > 0;`
282

100% 61:281

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 5	01:06:31	SELECT FIND_IN_SET('73', '16, 26, 97, 87')	1 row(s) returned	0.00073 sec / 0.0000...
✓ 6	01:11:04	SELECT FIND_IN_SET('87', '16, 26, 97, 87')	1 row(s) returned	0.00059 sec / 0.0000...
✓ 7	01:13:28	SELECT FIND_IN_SET('87', REPLACE('16, 26, 97, 87', ' ', ''))	1 row(s) returned	0.00088 sec / 0.0000...
✓ 8	01:19:27	CREATE TABLE transactions_products AS SELECT t.id AS transaction_id, p.id AS product_id FROM transactions_csv t JOIN pro...	253388 row(s) affected Records: 253388 Duplicates: 0... 1.396 sec	

Creamos la tabla aplicando el find_in_set con el replace para evitar problemas con el espacio después de las comas en las listas de los id de productos.

1 • SELECT * FROM ecommerce.transactions_products;

transaction_id	product...
00043A49-2948-494B-A5DD-A5BAE38B19DD	97
00043A49-2948-494B-A5DD-A5BAE38B19DD	97
00043A49-2948-494B-A5DD-A5BAE38B19DD	26
00043A49-2948-494B-A5DD-A5BAE38B19DD	16
00045D68-ED2E-4F2F-8186-CEE074D87500	81
00045D68-ED2E-4F2F-8186-CEE074D87500	30
00045D68-ED2E-4F2F-8186-CEE074D87500	16
00045D68-ED2E-4F2F-8186-CEE074D87500	11
00046103-1C2E-4FEF-83A0-AC0CE90A49B0	72
00051AA4-9C8E-4268-B070-C38062A1B3E2	18
0008A312-EDFE-4A4F-BC99-E9C92EC3CA4D	35
0008A312-EDFE-4A4F-BC99-E9C92EC3CA4D	33
0008A312-EDFE-4A4F-BC99-E9C92EC3CA4D	19
0009A151-9B0F-4E31-9053-A468F77FAAB	93
0009A151-9B0F-4E31-9053-A468F77FAAB	91
0009A151-9B0F-4E31-9053-A468F77FAAB	153

transactions_products 1

Action Output

Time	Action	Response	Duration / Fetch Time
01:11:04	SELECT FIND_IN_SET('87', '16, 26, 97, 87')	1 row(s) returned	0.00059 sec / 0.000...
01:13:28	SELECT FIND_IN_SET('87', REPLACE('16, 26, 97, 87', ','))	1 row(s) returned	0.00088 sec / 0.000...
01:19:27	CREATE TABLE transactions_products AS SELECT t.id AS transaction_id, p.id AS product_id FROM transactions_csv t JOIN pro...	253388 row(s) affected Records: 253388 Duplicates: 0...	1.396 sec
01:20:22	SELECT * FROM ecommerce.transactions_products	253388 row(s) returned	0.00080 sec / 0.065...

Tabla "transactions_products" creada.

👉 Necesitem conèixer el nombre de vegades que s'ha venut cada producte.

283 -- Necesitem conèixer el nombre de vegades que s'ha venut cada producte.

284

285

286 • SELECT pr.id,

287 pr.product_name,

288 COUNT(tp.product_id) AS purchase_amount

289 FROM products_csv pr

290 LEFT JOIN transactions_products tp

291 ON tp.product_id = pr.id

292 GROUP BY pr.id, pr.product_name

293 ORDER BY purchase_amount DESC;

294

id	product_name	purchase_amount
52	riverlands the duel	2654
26	Tully master Tatty	2635
21	duel Direwolf	2609
16	the duel warden	2608
66	mustafar jinn	2600
48	rock Renly in	2597
67	alith Jade	2597
33	duel warden	2597
23	riverlands north	2583
68	Stark Karstark	2580
88	Stannis warden s...	2587
28	chewbacca must...	2584
73	Dome bastard	2584
4	warden south duel	2584
51	Lannister	2580
34	of north	2580
38	pwok alith	2578
54	dooku solo	2575
36	Lannister	2575

Result 9

Action Output

Time	Action	Response	Duration / Fetch Time
01:20:19	SELECT pr.id, pr.product_name, COUNT(tp.product_id) AS purchase_amount FROM products_csv pr LEFT JOIN transactions_products tp	ON tp.product_id = pr...	100 row(s) retu... 0.173 sec / 0.000043...
01:26:29	SELECT pr.id, pr.product_name, COUNT(tp.product_id) AS purchase_amount FROM products_csv pr LEFT JOIN transactions_products tp	ON tp.product_id = pr...	100 row(s) retu... 0.173 sec / 0.000041...
01:28:47	SELECT pr.id, pr.product_name, COUNT(tp.product_id) AS purchase_amount FROM products_csv pr LEFT JOIN transactions_products tp	ON tp.product_id = pr...	100 row(s) retu... 0.172 sec / 0.000044...
01:31:22	SELECT pr.id, pr.product_name, COUNT(tp.product_id) AS purchase_amount FROM products_csv pr LEFT JOIN transactions_products tp	ON tp.product_id = pr...	100 row(s) retu... 0.173 sec / 0.000042...

Finalmente obtenemos la cantidad de veces que se ha vendido cada producto viendo que encabeza la lista el producto top ventas.